# Gaussian Natural Gradient Flows for Variational Inference

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## 1 Introduction

We consider the problem of sampling the target posterior distribution with density $\rho_{\text{post}}(\cdot)$, for the parameter $\theta \in \mathbb{R}^{N_\theta}$, given by

$$\rho_{\text{post}}(\theta) \propto \exp(-V(\theta)), \qquad \theta \in \mathbb{R}^{N_\theta},$$

where $V(\theta) : \mathbb{R}^{N_\theta} \to \mathbb{R}$ is the known function while the normalizing constant $Z = \int \exp(-V(\theta))\mathrm{d}\theta$ is untractable. Markov chain Monte Carlo (MCMC) methods [5] aim to sample from target distribution $\rho_{\text{post}}$ asymptotically, but can be computationally expensive in high-dimensional or ill-conditional settings [14]. Variational Inference (VI) [4], on the other hand, replaces exact sampling by optimization over a tractable family of probability distributions on its Kullback-Leibler (KL) divergence to the true target posterior, which is often more scalable than MCMC [9].

We focus on the Gaussian variational family

$$\mathcal{P}_{\text{G}} := \left\{\rho_a = \mathcal{N}(m, C) : a = (m, C) \in \mathcal{A} := \mathbb{R}^{N_\theta} \times \mathbb{S}_{++}(N_\theta)\right\},$$

which is the simplest nontrivial VI family, where the Gaussian variational inference problem is

$$a_\star = (m_\star, C_\star) \in \operatorname*{argmin}_{a\in\mathcal{A}} \mathcal{E}(a), \qquad \mathcal{E}(a) := \mathrm{KL}(\rho_a \,\|\, \rho_{\text{post}}).$$

Recent theoretical work has studied Gaussian variational inference through Wasserstein and Bures–Wasserstein geometry [8, 10]. This viewpoint connects Gaussian variational inference with optimal transport and constrained Wasserstein gradient flows, which shows excellent convergence theory for both Wasserstein gradient flow and forward-backward algorithms. On the other side, Fisher–Rao gradient flow [6] develops a complementary information-geometric viewpoint [2]. When the Fisher–Rao gradient flow is restricted to the Gaussian family, we obtain the natural gradient flow [1] for Gaussian variational inference [7]. In addition, Fisher–Rao gradient flow is affine invariant [7], which is expected to perform better than non-affine-invariant methods like Wasserstein gradient flow at ill-conditioned initiation. However, while natural gradient flow is useful in practice, the convergence theory for Gaussian variational inference remains underexplored. We aim to develop the convergence theory of Gaussian natural gradient flows of form

$$\begin{aligned}
\frac{\mathrm{d}m_t}{\mathrm{d}t} &= C_t \mathbb{E}_{\rho_{a_t}}[\nabla_\theta \log \rho_{\text{post}}(\theta)], \\
\frac{\mathrm{d}C_t}{\mathrm{d}t} &= C_t + C_t \mathbb{E}_{\rho_{a_t}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_t.
\end{aligned} \tag{1}$$

## 2 Global convergence and discretization

We study the Gaussian natural gradient flow where the target posterior distribution $\rho_{\text{post}}$ is logconcave, and then propose practical discretization algorithms.

**Assumption 2.1.** We assume that $\rho_{\text{post}}$ is $\alpha$-strongly logconcave and $\beta$-smooth, i.e.

$$\alpha I \preceq -\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta) \preceq \beta I,$$

which is equivalent to

$$\alpha I \preceq \nabla^2 V(\theta) \preceq \beta I.$$

We first show the spectral bounds the covariance along (1) under Assumption 2.1.

**Lemma 2.2.** *Under Assumption 2.1, if the initial covariance satisfies* $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$*, then along* (1)*,* $\lambda_{\min} I \preceq C_t \preceq \lambda_{\max} I$ *with*

$$\lambda_{\min} = \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}, \qquad \lambda_{\max} = \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}.$$

*Proof.* We parameterize the eigenvalues of $C_t$ as $\lambda_1(t), \lambda_2(t), \ldots, \lambda_{N_\theta}(t)$ with corresponding normalized eigenvectors $u_1(t), u_2(t), \ldots, u_{N_\theta}(t)$. Hence, we have the ODEs of the eigenvalues:

$$\frac{\mathrm{d}\lambda_i(t)}{\mathrm{d}t} = u_i(t)^\top \frac{\mathrm{d}C_t}{\mathrm{d}t} u_i(t).$$

For the Gaussian natural flow (1), since $\alpha I \preceq -\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta) \preceq \beta I$, then

$$C_t - \beta C_t^2 \preceq \frac{\mathrm{d}C_t}{\mathrm{d}t} = C_t + C_t \mathbb{E}_{\rho_{a_t}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_t \preceq C_t - \alpha C_t^2.$$

We thus obtain the ODE of eigenvalues

$$\lambda_i(t) - \beta \lambda_i(t)^2 \leq \frac{\mathrm{d}\lambda_i(t)}{\mathrm{d}t} \leq \lambda_i(t) - \alpha \lambda_i(t)^2,$$

with initialization condition $\lambda_{0,\min} \leq \lambda_i(0) \leq \lambda_{0,\max}$.

If $\lambda_i(t) \geq 1/\alpha$, then $\frac{\mathrm{d}\lambda_i(t)}{\mathrm{d}t} \leq 0$; if $\lambda_i(t) \leq 1/\beta$, then $\frac{\mathrm{d}\lambda_i(t)}{\mathrm{d}t} \geq 0$, it follows that $\min\{\lambda_{0,\min}, 1/\beta\} \leq \lambda_i(t) \leq \max\{\lambda_{0,\max}, 1/\alpha\}$, hence $\lambda_{\min} I \preceq C_t \preceq \lambda_{\max} I$. □

**Theorem 2.3.** *Under Assumption 2.1, if the initial covariance has spectral bounds* $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0\,\max} I$*, then for the Gaussian natural gradient flow* (1)*, we have the following global convergence guarantee*

$$\mathcal{E}(a_t) - \mathcal{E}(a_\star) \leq e^{-2\alpha\lambda_{\min} t}(\mathcal{E}(a_0) - \mathcal{E}(a_\star)),$$

*where* $\lambda_{\min} = \min\{\lambda_{0,\min}, 1/\beta\}$ *as in Lemma 2.2.*

*Proof.* Since $\mathcal{E}$ is $\alpha$-strongly convex on the entire Wasserstein space $(\mathcal{P}_2(\mathbb{R}^{N_\theta}), W_2)$, then the energy functional is $\alpha$-strongly convex along $W_2$-geodesics [10], we thus obtain the Polyak–Łojasiewicz (PL) type inequality on Gaussians:

$$\begin{aligned}
\|\operatorname{grad}^{W_2} \mathcal{E}(a_t)\|_{W_2}^2 = &\|\mathbb{E}_{\rho_{a_t}}[\nabla_\theta \log \rho_{\text{post}}(\theta)]\|^2 \\
&+ \operatorname{Tr}((C_t^{-1} + \mathbb{E}_{\rho_{a_t}}[\nabla_\theta \nabla_\theta \log_{\text{post}}(\theta)]) C_t (C_t^{-1} + \mathbb{E}_{\rho_{a_t}}[\nabla_\theta \nabla_\theta \log_{\text{post}}(\theta)])) \\
\geq &\, 2\alpha(\mathcal{E}(a_t) - \mathcal{E}(a_\star)).
\end{aligned}$$

For the Gaussian natural gradient flow (1), we compute the gradient norm in Fisher–Rao geometry and connect it to the Wasserstein geometry:

$$\begin{aligned}\|\operatorname{grad}\mathcal{E}(a_t)\|_{a_t}^2 &= (\mathbb{E}_{\rho_{a_t}}[\nabla_\theta \log \rho_{\mathrm{post}}(\theta)])^\top C_t(\mathbb{E}_{\rho_{a_t}}[\nabla_\theta \log \rho_{\mathrm{post}}(\theta)]) \\ &\quad + \mathrm{Tr}(C_t(C_t^{-1} + \mathbb{E}_{\rho_{a_t}}[\nabla_\theta\nabla_\theta \log_{\mathrm{post}}(\theta)])C_t(C_t^{-1} + \mathbb{E}_{\rho_{a_t}}[\nabla_\theta\nabla_\theta \log_{\mathrm{post}}(\theta)])) \\ &\geq \lambda_{\min}\|\operatorname{grad}^{W_2}\mathcal{E}(a_t)\|_{W_2}^2 \geq 2\alpha\lambda_{\min}(\mathcal{E}(a_t) - \mathcal{E}(a_\star)),\end{aligned}$$

where we have applied the spectral bound $\lambda_{\min} I \preceq C_t$ from Lemma 2.2. This gives

$$\frac{\mathrm{d}\mathcal{E}(a_t)}{\mathrm{d}t} \leq -2\alpha\lambda_{\min}(\mathcal{E}(a_t) - \mathcal{E}(a_\star)),$$

by Gronwall's inequality,

$$\mathcal{E}(a_t) - \mathcal{E}(a_\star) \leq e^{-2\alpha\lambda_{\min} t}(\mathcal{E}(a_0) - \mathcal{E}(a_\star)).$$

□

### 2.1 Discretization with Riemannian distance

We decompose the objective as

$$\mathcal{E}(a) = \mathrm{KL}(\rho_a \| \rho_{\mathrm{post}}) = \mathcal{E}_1(a) + \mathcal{E}_2(a),$$

where

$$\mathcal{E}_1(a) = \int \rho_a \log \rho_a = -\frac{1}{2}\log\det C + \mathrm{const.}, \qquad \mathcal{E}_2(a) = -\int \rho_a \log \rho_{\mathrm{post}} = -\mathbb{E}_{\rho_a}[\log \rho_{\mathrm{post}}].$$

As the Fisher–Rao metric induced on the Gaussian family reads

$$g_a^{\mathrm{FR}}((u, X), (v, Y)) = u^\top C^{-1} v + \frac{1}{2}\mathrm{Tr}(C^{-1} X C^{-1} Y),$$

then we define the squared Riemannian distance at local iterate $a_n = (m_n, C_n)$ as

$$d(a, a_n)^2 = (m - m_n)^\top C_n^{-1}(m - m_n) + \frac{1}{2}\|Z\|_F^2, \qquad Z = \log(C_n^{-1/2} C C_n^{-1/2}).$$

As the entropy term $\mathcal{E}_1$ is often non-smooth, we apply proximal algorithm [11] for optimization. In the forward step,

$$a_{n+1/2} = \underset{a \in \mathcal{A}}{\operatorname{argmin}} \left\{ \mathcal{E}_2(a_n) + \langle \operatorname{grad}\mathcal{E}_2(a_n), \mathrm{Log}_{a_n}(a) \rangle + \frac{1}{2\Delta t} d(a, a_n)^2 \right\},$$

which gives

$$\begin{aligned} m_{n+1/2} &= m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\mathrm{post}}(\theta)], \\ C_{n+1/2} &= C_n^{1/2}\exp(\Delta t C_n^{1/2}\mathbb{E}_{\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)] C_n^{1/2}) C_n^{1/2}. \end{aligned}$$

The backward step reads

$$a_{n+1} = \underset{a \in \mathcal{A}}{\operatorname{argmin}} \left\{ \mathcal{E}_1(a) + \frac{1}{2\Delta t} d(a, a_{n+1/2})^2 \right\},$$

which gives

$$m_{n+1} = m_{n+1/2}, \qquad C_{n+1} = e^{\Delta t} C_{n+1/2}.$$

These collectively give the update through Riemannian geometry:

$$\begin{aligned} m_{n+1} &= m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)], \\ C_{n+1} &= e^{\Delta t} C_n^{1/2} \exp(\Delta t C_n^{1/2} \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2}) C_n^{1/2}. \end{aligned} \tag{2}$$

We proceed to show the convergence theory of (2) under Assumption 2.1. We first show that the update for each step is exactly the geodesic gradient descent on the Gaussian manifold.

**Proposition 2.4.** *The discrete algorithm with Riemannian distance* (2) *gives the following one-step update:*

$$a_{n+1} = \operatorname{Exp}_{a_n}(-\Delta t \operatorname{grad} \mathcal{E}(a_n)).$$

*Proof.* The Riemannian gradient of $\mathcal{E}(a_n)$ reads

$$\operatorname{grad} \mathcal{E}(a_n) = (-C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)], -C_n - C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n).$$

For $a = (m, C) \in \mathcal{A}$ and $(u, U) \in T_a\mathcal{A}$, the exponential map on the Gaussian manifold reads

$$\operatorname{Exp}_{(m,C)}(u, U) = \left(m + u, C^{1/2} \exp(C^{-1/2} U C^{-1/2}) C^{1/2}\right).$$

In the mean direction, we have

$$m_{n+1} = m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)] = m_n - \Delta t \operatorname{grad}_m \mathcal{E}(a_n).$$

In the covariance direction,

$$C_{n+1} = C_n^{1/2} \exp(\Delta t I + \Delta t C_n^{1/2} \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2}) C_n^{1/2},$$

which is exactly the covariance component of $\operatorname{Exp}_{a_n}(-\Delta t \operatorname{grad} \mathcal{E}(a_n))$ because

$$\begin{aligned} -\Delta t \operatorname{grad}_C \mathcal{E}(a_n) &= \Delta t C_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n \\ &= C_n^{1/2} (\Delta t I + \Delta t C_n^{1/2} \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2}) C_n^{1/2}, \end{aligned}$$

which completes the proof. □

We then prove the spectral bounds of covariance along (2).

**Lemma 2.5.** *Under Assumption 2.1 and assume that the initial covariance has spectral bounds* $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$, *then along the discrete algorithm with Riemannian distance* (2), *if the stepsize is chosen to be* $0 < \Delta t \leq \frac{1}{\beta \lambda_{\max}}$, *one has*

$$\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I,$$

*where we denote* $\lambda_{\min} := \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}$ *and* $\lambda_{\max} := \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}$.

*Proof.* We proceed with induction, it is known that $\lambda_{\min} I \preceq C_0 \preceq \lambda_{\max} I$. We then assume that $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$. Since $\alpha I \preceq -\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta) \preceq \beta I$, then

$$\exp\left(-\Delta t \beta C_n\right) \preceq \exp\left(\Delta t C_n^{1/2} \mathbb{E}_{\theta \sim \rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2}\right) \preceq \exp\left(-\Delta t \alpha C_n\right),$$

and hence

$$C_n \exp\left(\Delta t(I - \beta C_n)\right) \preceq C_{n+1} \preceq C_n \exp\left(\Delta t(I - \alpha C_n)\right).$$

For the upper bound, we consider $f(x) = x \exp(\Delta t(1 - \alpha x))$, when $\Delta t \le \frac{1}{\beta \lambda_{\max}}$, $f'(x) > 0$ on $x \in [\lambda_{\min}, \lambda_{\max}]$, hence

$$C_{n+1} \preceq f(\lambda_{\max}) I = \lambda_{\max} \exp\left(\Delta t(1 - \alpha \lambda_{\max})\right) I \preceq \lambda_{\max} I.$$

For the lower bound, we consider $g(x) = x \exp(\Delta t(1 - \beta x))$. Since $\Delta t \le \frac{1}{\beta \lambda_{\max}}$, then $g'(x) > 0$ on $x \in [\lambda_{\min}, \lambda_{\max}]$, hence

$$C_{n+1} \succeq g(\lambda_{\min}) I = \lambda_{\min} \exp\left(\Delta t(1 - \beta \lambda_{\min})\right) \succeq \lambda_{\min} I.$$

Hence $\lambda_{\min} I \preceq C_{n+1} \preceq \lambda_{\max} I$, which completes the proof. □

We proceed to prove the global convergence rate of (2).

**Theorem 2.6.** *Under Assumption 2.1 with the covariance along* (2) *bounded by* $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$ *(Lemma 2.5), then if the stepsize satisfies* $0 < \Delta t \le \frac{1}{\beta \lambda_{\max}}$*,* (2) *gives the following global convergence rate:*

$$\mathcal{E}(a_N) - \mathcal{E}(a_\star) \le (1 - \alpha \lambda_{\min} \Delta t(2 - \beta \lambda_{\max} \Delta t))^N (\mathcal{E}(a_0) - \mathcal{E}(a_\star)).$$

*Proof.* We first prove the smoothness of $\mathcal{E}$ in the Riemannian geometry. For a fixed $a = (m, C) \in \mathcal{A}$, we write $C = BB^\top$. Let $\zeta = (u, U) \in T_a \mathcal{A}$, we denote $U = BSB$, where $S = S^\top$, the corresponding exponential curve can be parameterized by

$$m_t = m + tu, \qquad C_t = Be^{tS} B.$$

We also write $B_t = Be^{tS/2}$, $X_t = m_t + B_t \xi$, where $\xi \sim \mathcal{N}(0, I)$ so that $X_t \sim \rho_{a_t}$. Define

$$\psi(t) := \mathcal{E}_2(a_t) = \mathbb{E}[V(X_t)], \qquad V(\theta) := -\log \rho_{\text{post}}(\theta).$$

We compute

$$\frac{\mathrm{d}X_t}{\mathrm{d}t} = u + \frac{1}{2} B_t S \xi, \qquad \frac{\mathrm{d}^2 X_t}{\mathrm{d}t^2} = \frac{1}{4} B_t S^2 \xi,$$

and $\alpha I \preceq \nabla^2 V \preceq \beta I$, then

$$\begin{aligned}
\psi''(t) &= \mathbb{E}\left[\left(\frac{\mathrm{d}X_t}{\mathrm{d}t}\right)^\top \nabla^2 V(X_t) \left(\frac{\mathrm{d}X_t}{\mathrm{d}t}\right)\right] + \mathbb{E}\left[\nabla V(X_t)^\top \left(\frac{\mathrm{d}^2 X_t}{\mathrm{d}t^2}\right)\right] \\
&\le \beta \mathbb{E}\left\|\frac{\mathrm{d}X_t}{\mathrm{d}t}\right\|^2 + \frac{1}{4}\mathbb{E}\left[\operatorname{Tr}\left(S^2 B_t^\top \nabla^2 V(X_t) B_t\right)\right] \qquad \text{(by Stein's Lemma)} \\
&\le \beta\left(\|u\|^2 + \frac{1}{4}\|B_t S\|_F^2\right) + \frac{\beta}{4}\|B_t S\|_F^2 = \beta \|u\|^2 + \frac{\beta}{2}\|B_t S\|_F^2 \\
&\le \beta \lambda_{\max} u^\top C^{-1} u + \frac{\beta}{2} \lambda_{\max} \|S\|_F^2.
\end{aligned}$$

Since the tangent norm is $\|\zeta\|_a^2 = u^\top C^{-1} u + \frac{1}{2}\|S\|_F^2$, then

$$\psi''(t) \le L\|\zeta\|_a^2, \qquad L = \beta\lambda_{\max}.$$

We apply Taylor's expansion along the exponential curve to obtain the $L$-smoothness of $\mathcal{E}_2$:

$$\mathcal{E}_2(\mathrm{Exp}_a(\zeta)) \le \mathcal{E}_2(a) + \langle \mathrm{grad}\mathcal{E}_2(a), \zeta\rangle_a + \frac{L}{2}\|\zeta\|_a^2.$$

We also note that

$$\log\det C_t = \log\det C + t\,\mathrm{Tr}(S),$$

hence $\mathcal{E}_1$ is geodesically affine, these collectively give the $L$-smoothness of the energy functional $\mathcal{E}$:

$$\mathcal{E}(\mathrm{Exp}_a(\zeta)) \le \mathcal{E}(a) + \langle \mathrm{grad}\mathcal{E}(a), \zeta\rangle_a + \frac{L}{2}\|\zeta\|_a^2. \tag{3}$$

Since by Proposition 2.4 we have

$$a_{n+1} = \mathrm{Exp}_{a_n}\left(-\Delta t\,\mathrm{grad}\,\mathcal{E}(a_n)\right),$$

then we choose $a = a_n$ and $\zeta = -\Delta t\,\mathrm{grad}\,\mathcal{E}(a_n)$ in (3) to obtain

$$\begin{aligned}
\mathcal{E}(a_{n+1}) &\le \mathcal{E}(a_n) + \langle \mathrm{grad}\mathcal{E}(a_n), -\Delta t\,\mathrm{grad}\,\mathcal{E}(a_n)\rangle_{a_n} + \frac{L}{2}\left\|-\Delta t\,\mathrm{grad}\,\mathcal{E}(a_n)\right\|_{a_n}^2 \\
&= \mathcal{E}(a_n) - \Delta t\left(1 - \frac{L\Delta t}{2}\right)\|\mathrm{grad}\mathcal{E}(a_n)\|_{a_n}^2.
\end{aligned}$$

We connect the affine invariant geometry to Wasserstein geometry as in the proof of Theorem 2.3, which gives

$$\|\mathrm{grad}\mathcal{E}(a_n)\|_{a_n}^2 \ge \lambda_{\min}\left\|\mathrm{grad}^{W_2}\mathcal{E}(a_n)\right\|_{W_2}^2 \ge 2\alpha\lambda_{\min}(\mathcal{E}(a_n) - \mathcal{E}(a_\star)).$$

We therefore get

$$\begin{aligned}
\mathcal{E}(a_{n+1}) - \mathcal{E}(a_\star) &\le \mathcal{E}(a_n) - \mathcal{E}(a_\star) - \Delta t\left(1 - \frac{L\Delta t}{2}\right)\|\mathrm{grad}\mathcal{E}(a_n)\|_{a_n}^2 \\
&\le (1 - \alpha\lambda_{\min}\Delta t(2 - L\Delta t))(\mathcal{E}(a_n) - \mathcal{E}(a_\star)),
\end{aligned}$$

as $\Delta t \le \frac{1}{L} < \frac{2}{L}$, we have the following convergence guarantee:

$$\mathcal{E}(a_N) - \mathcal{E}(a_\star) \le (1 - \alpha\lambda_{\min}\Delta t(2 - L\Delta t))^N(\mathcal{E}(a_0) - \mathcal{E}(a_\star)).$$

□

### 2.2 Discretization with KL divergence

The discrete algorithm with Riemannian distance (2) is geometrically natural but computationally expensive, which requires to compute matrix exponential at every step, for the ease in computation, we develop the following discretization scheme with KL divergence.

At the forward step, we compute

$$a_{n+1/2} = \operatorname*{argmin}_{a\in\mathcal{A}}\left\{\mathcal{E}_2(a_n) + \langle\nabla\mathcal{E}_2(a_n), a - a_n\rangle + \frac{1}{\Delta t}\mathrm{KL}(\rho_a\|\rho_{a_n})\right\},$$

which gives

$$\begin{aligned}
m_{n+1/2} &= m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)], \\
C_{n+1/2} &= (C_n^{-1} - \Delta t \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)])^{-1}.
\end{aligned}$$

For the backward step, the update reads

$$a_{n+1} = \underset{a \in \mathcal{A}}{\operatorname{argmin}} \left\{ \mathcal{E}_1(a) + \frac{1}{\Delta t} \mathrm{KL}(\rho_a \| \rho_{a_{n+1/2}}) \right\},$$

which gives

$$m_{n+1} = m_{n+1/2}, \qquad C_{n+1} = (1 + \Delta t) C_{n+1/2}.$$

Hence, the discrete algorithm with KL divergence reads

$$\begin{aligned}
m_{n+1} &= m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)], \\
C_{n+1} &= (1 + \Delta t)(C_n^{-1} - \Delta t \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)])^{-1}.
\end{aligned} \tag{4}$$

We denote $D(a, a_n) = \mathrm{KL}(\rho_a \| \rho_{a_n})$ and take note that $D(a, a_n)$ is a Bregman divergence because

$$D_\psi(\xi, \xi_n) = \psi(\xi) - \psi(\xi_n) - \langle \nabla \psi(\xi_n), \xi - \xi_n \rangle = \mathrm{KL}(\rho_a \| \rho_{a_n}),$$

where we define

$$\psi(\xi) = -\frac{1}{2} \log \det(\xi_2 - \xi_1 \xi_1^\top), \qquad \xi = (\xi_1, \xi_2) = (m, C + mm^\top).$$

We proceed to prove the convergence rate of discretization algorithm with KL divergence (4), under the standard Assumption 2.1. We first prove the spectral bounds of covariance along (4).

**Lemma 2.7.** *Under Assumption 2.1, if $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$, then along the discrete update (4), the covariance $C_n$ satisfies $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$, where*

$$\lambda_{\min} = \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}, \qquad \lambda_{\max} = \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}.$$

*Proof.* We re-write the covariance update in (4) as

$$C_{n+1}^{-1} = \frac{1}{1 + \Delta t}(C_n^{-1} - \Delta t \mathbb{E}_{\theta \sim \rho_{a_n}}[\nabla_\theta^2 \log \rho_{\text{post}}(\theta)]).$$

As we have $\alpha I \preceq -\mathbb{E}_{\theta \sim \rho_{a_n}}[\nabla_\theta^2 \log \rho_{\text{post}}(\theta)] \preceq \beta I$, for $\alpha \le h \le \beta$, the update in eigenvalue is given by

$$\lambda_{\text{new}} = \left( \frac{1}{1 + \Delta t} \left( \frac{1}{\lambda} + \Delta t h \right) \right)^{-1} = \frac{1 + \Delta t}{\frac{1}{\lambda} + \Delta t h}.$$

Hence if $\lambda_{\min} \le \lambda \le \lambda_{\max}$, the update is bounded by

$$\lambda_{\min} \le \frac{1 + \Delta t}{\frac{1}{\lambda_{\min}} + \Delta t \beta} \le \lambda_{\text{new}} \le \frac{1 + \Delta t}{\frac{1}{\lambda_{\max}} + \Delta t \alpha} \le \lambda_{\max},$$

where the first and the last inequalities follow from the fact that $\lambda_{\min} \le 1/\beta$ and $\lambda_{\max} \ge 1/\alpha$. Hence the eigenvalues of $C_n$ is bounded by mathematical induction, i.e., $\lambda_{\min} I \preceq C_0 \preceq \lambda_{\max} I$ and if $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$, then $\lambda_{\min} I \preceq C_{n+1} \preceq \lambda_{\max} I$. $\square$

**Proposition 2.8.** *Under Assumption 2.1 with covariance bound $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$ (Lemma 2.7) along (4), $\mathcal{E}_2$ is L-smooth with respect to local Bregman divergence $D(\cdot, a_n)$, i.e., for any $a = (m, C) \in \mathcal{A}$ with $\lambda_{\min} I \preceq C \preceq \lambda_{\max} I$, one has*

$$\mathcal{E}_2(a) \leq \mathcal{E}_2(a_n) + \langle \nabla \mathcal{E}_2(a_n), a - a_n \rangle + LD(a, a_n),$$

*for a fixed constant $L = \beta \lambda_{\max} \max\left\{1, \frac{\lambda_{\max}^3}{2\lambda_{\min}^3}\right\}$.*

*Proof.* We consider the displacement from $(m_0, C_0) \in \mathcal{A}$ to $(m, C) \in \mathcal{A}$, and denote the direction $u = m - m_0$ and $H = B - B_0$ with $C = BB^\top$ and $C_0 = B_0 B_0^\top$. Under this formulation, we write $\mathcal{E}_2$ as

$$\mathcal{E}_2(m, B) := \mathbb{E}_{\xi \sim \mathcal{N}(0, I_{N_\theta})}[V(m + B\xi)].$$

We denote the trajectory

$$\varphi(t) = \mathbb{E}_{\xi \sim \mathcal{N}(0, I_{N_\theta})}[V(m_0 + tu + (B_0 + tH)\xi)].$$

We write $X_t = m_t + B_t \xi$, and the first and second derivatives of $\varphi$ read

$$\varphi'(t) = \mathbb{E}_{\xi \sim \mathcal{N}(0, I_{N_\theta})}[\nabla V(X_t)(u + H\xi)], \qquad \varphi''(t) = \mathbb{E}_{\xi \sim \mathcal{N}(0, I_{N_\theta})}[(u + H\xi)^\top \nabla^2 V(X_t)(u + H\xi)].$$

As we assume $\nabla^2 V \preceq \beta I$, then

$$\varphi''(t) \leq \beta(\|u\|^2 + \|H\|_F^2).$$

As the Taylor's expansion of $\varphi(t)$ gives

$$\varphi(1) = \varphi(0) + \varphi'(0) + \int_0^1 (1 - s)\varphi''(s) \mathrm{d}s,$$

then

$$\varphi(1) - \varphi(0) - \varphi'(0) \leq \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2),$$

which is equivalent to

$$\mathcal{E}_2(a) - \mathcal{E}_2(a_0) - \langle \nabla_m \mathcal{E}_2(a_0), u \rangle - \langle \nabla_B \mathcal{E}_2(a_0), H \rangle \leq \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2).$$

We denote the gradient of $\mathcal{E}_2$ with respect to $C$ as

$$G_0 := \nabla_C \mathcal{E}_2(a_0) = \frac{1}{2}\mathbb{E}_{\theta \sim \mathcal{N}(m_0, C_0)}[\nabla_\theta^2 V(\theta)],$$

where we have $G_0 \succeq 0$ as $V$ is $\alpha$-strongly convex. We also compute the gradient of $\mathcal{E}_2$ with respect to $B$:

$$\nabla_B \mathcal{E}_2(a_0) = \nabla_C \mathcal{E}_2(a_0) \cdot 2B_0 = 2G_0 B_0,$$

this gives

$$\begin{aligned}
\langle \nabla_B \mathcal{E}_2(a_0), H \rangle &= \mathrm{Tr}((2G_0 B_0)^\top H) = \mathrm{Tr}(G_0(B_0 H + HB_0)) \\
&\leq \mathrm{Tr}(G_0(B_0 H + HB_0 + H^2)) = \mathrm{Tr}(G_0^\top((B_0 + H)(B_0 + H)^\top - B_0 B_0^\top)) \\
&= \langle \nabla_C \mathcal{E}_2(a_0), C - C_0 \rangle,
\end{aligned}$$

where the inequality comes from the fact that $G_0 \succeq 0$ and $H^2 \succeq 0$.

We therefore obtain

$$\begin{aligned}
\mathcal{E}_2(a) &\leq \mathcal{E}_2(a_0) + \langle \nabla_m \mathcal{E}_2(a_0), m - m_0 \rangle + \langle \nabla_C \mathcal{E}_2(a_0), C - C_0 \rangle + \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2) \\
&= \mathcal{E}_2(a_0) + \langle \nabla \mathcal{E}_2(a_0), a - a_0 \rangle + \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2).
\end{aligned}$$

We then connect the divergence term $\frac{\beta}{2}(\|u\|^2 + \|H\|_F^2)$ to the KL divergence $D$. For the mean part,

$$\frac{\beta}{2}\|u\|^2 \leq \beta \lambda_{\max} \cdot \frac{1}{2} u^\top C_n^{-1} u \leq \beta \lambda_{\max} D(a, a_n).$$

For the covariance part, we denote $M := C_n^{-1/2} C C_n^{-1/2}$, and hence the divergence for the covariance part reads $D^C(C, C_n) = \frac{1}{2}(\mathrm{Tr}(M) - \log\det M - N_\theta)$, and the eigenvalues $\eta_1, \ldots \eta_{N_\theta}$ of $M$ satisfy $\eta_i \in \left[\frac{\lambda_{\min}}{\lambda_{\max}}, \frac{\lambda_{\max}}{\lambda_{\min}}\right]$.

We denote $\phi(\eta) = \eta - \log \eta - 1$, and conduct Taylor expansion at $\eta = 1$ with $\zeta \in \left[\frac{\lambda_{\min}}{\lambda_{\max}}, \frac{\lambda_{\max}}{\lambda_{\min}}\right]$:

$$\phi(\eta) = \frac{1}{2}\phi''(\zeta)(\eta - 1)^2 = \frac{1}{2\zeta^2}(\eta - 1)^2 \geq \frac{\lambda_{\min}^2}{2\lambda_{\max}^2}(\eta - 1)^2,$$

this gives the following bound:

$$\begin{aligned}
D^C(C, C_n) &= \frac{1}{2}\sum_{i=1}^{N_\theta} \phi(\eta_i) \geq \frac{\lambda_{\min}^2}{4\lambda_{\max}^2}\sum_{i=1}^{N_\theta}(\eta_i - 1)^2 = \frac{\lambda_{\min}^2}{4\lambda_{\max}^2}\|M - I\|_F^2 \\
&\geq \frac{\lambda_{\min}^2}{4\lambda_{\max}^2} \cdot \frac{1}{\lambda_{\max}^2} \cdot \|C - C_n\|_F^2 = \frac{\lambda_{\min}^2}{4\lambda_{\max}^4}\|B_n H + H B_n\|_F^2 \geq \frac{\lambda_{\min}^3}{\lambda_{\max}^4}\|H\|_F^2.
\end{aligned}$$

These collectively give,

$$\frac{\beta}{2}(\|u\|^2 + \|H\|_F^2) \leq L D(a, a_n), \qquad L = \beta \lambda_{\max} \max\left\{1, \frac{\lambda_{\max}^3}{2\lambda_{\min}^3}\right\},$$

which gives

$$\mathcal{E}_2(a) \leq \mathcal{E}_2(a_n) + \langle \nabla \mathcal{E}_2(a_n), a - a_n \rangle + L D(a, a_n).$$

□

**Lemma 2.9** (One-step inequality)**.** *Under Assumption 2.1, for the discrete update (4) with stepsize $0 < \Delta t \leq \frac{1}{L}$ with $L$ as in Proposition 2.8, we have the following one-step inequality:*

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t} D(a_n, a_{n+1}).$$

*Proof.* As the discrete update is characterized by

$$a_{n+1} = \operatorname*{argmin}_{a \in \mathcal{A}} \left\{ \mathcal{E}_1(a) + \langle \nabla \mathcal{E}_2(a_n), a - a_n \rangle + \frac{1}{\Delta t} D(a, a_n) \right\},$$

then the first order condition gives that, there exists $g_{n+1} \in \partial \mathcal{E}_1(a_{n+1})$, such that

$$g_{n+1} + \nabla \mathcal{E}_2(a_n) + \frac{1}{\Delta t}(\nabla \psi(\xi_{n+1}) - \nabla \psi(\xi_n)) = 0.$$

We take $\langle \cdot, a_n - a_{n+1}\rangle$, which gives

$$\langle g_{n+1}, a_n - a_{n+1}\rangle + \langle \nabla \mathcal{E}_2(a_n), a_n - a_{n+1}\rangle + \frac{1}{\Delta t}\langle \nabla\psi(\xi_{n+1}) - \nabla\psi(\xi_n), a_n - a_{n+1}\rangle = 0.$$

Since $\mathcal{E}_1 = -\frac{1}{2}\log\det C + \text{const.}$ is convex, then $\mathcal{E}_1(a_n) \geq \mathcal{E}_1(a_{n+1}) + \langle g_{n+1}, a_n - a_{n+1}\rangle$, hence

$$\langle \nabla \mathcal{E}_2(a_n), a_{n+1} - a_n\rangle \leq \mathcal{E}_1(a_n) - \mathcal{E}_1(a_{n+1}) + \frac{1}{\Delta t}\langle \nabla\psi(\xi_{n+1}) - \nabla\psi(\xi_n), a_n - a_{n+1}\rangle.$$

We then apply the three-points lemma [3, Lemma 9.11] to obtain

$$\langle \nabla \mathcal{E}_2(a_n), a_{n+1} - a_n\rangle \leq \mathcal{E}_1(a_n) - \mathcal{E}_1(a_{n+1}) + \frac{1}{\Delta t}(D(a_n, a_n) - D(a_n, a_{n+1}) - D(a_{n+1}, a_n)). \tag{5}$$

By Proposition 2.8, the relative smoothness of $\mathcal{E}_2$ gives

$$\mathcal{E}_2(a_{n+1}) \leq \mathcal{E}_2(a_n) + \langle \nabla \mathcal{E}_2, a_{n+1} - a_n\rangle + LD(a_{n+1}, a_n).$$

We take the summation of the above inequality and (5), this gives

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t}D(a_n, a_{n+1}) - \left(\frac{1}{\Delta t} - L\right)D(a_{n+1}, a_n),$$

if we furthermore choose stepsize $0 < \Delta t \leq \frac{1}{L}$, then we obtain the following one-step inequality:

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t}D(a_n, a_{n+1}).$$

$\square$

We proceed to show the convergence of (4) by bounding the Bregman/KL divergence term.

**Theorem 2.10.** *Under Assumption 2.1 with spectral bounds $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$ (Lemma 2.7) along (4), for stepsize $0 < \Delta t \leq \frac{1}{L}$ with $L$ as in Proposition 2.8, (4) yields the following convergence guarantee:*

$$\mathcal{E}(a_N) - \mathcal{E}(a_\star) \leq (1 - 2\alpha\cdot\kappa(\Delta t)\Delta t)^N(\mathcal{E}(a_0) - \mathcal{E}(a_\star)),$$

*where*

$$\kappa(\Delta t) = \lambda_{\min}\min\left\{\frac{1 + \Delta t\alpha\lambda_{\min}}{2(1+\Delta t)}, \frac{1}{4(1+\Delta t\beta\lambda_{\max})^2}\right\}.$$

*Proof.* We first connect $D$ to the Wasserstein gradient norm $\|\operatorname{grad}^{W_2}\mathcal{E}(a_n)\|_{W_2}^2$. We denote that

$$g_n := \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)], \quad H_n := \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)], \quad A_n = C_n^{-1} + H_n.$$

For the mean part, from (4), we have

$$\begin{aligned} D^m(a_n, a_{n+1}) &= \frac{1}{2}(m_n - m_{n+1})^\top C_{n+1}^{-1}(m_n - m_{n+1}) = \frac{\Delta t^2}{2}g_n^\top C_n C_{n+1} C_n g_n \\ &= \frac{\Delta t^2}{2(1+\Delta t)}(g_n^\top C_n g_n - \Delta t g_n^\top C_n H_n C_n g_n) \geq \frac{\Delta t^2\lambda_{\min}}{2(1+\Delta t)}(1 + \Delta t\alpha\lambda_{\min})\|g_n\|^2, \end{aligned}$$

where the inequality comes from $-H_n \succeq \alpha I$ and $C_n \succeq \lambda_{\min} I$ (Lemma 2.7).

For the covariance part, we have

$$D^C(C_n, C_{n+1}) = \frac{1}{2}(\mathrm{Tr}(C_{n+1}^{-1}C_n) - \log\det(C_{n+1}^{-1}C_n) - N_\theta).$$

From (4),

$$C_{n+1}^{-1}C_n = \frac{1}{1+\Delta t}(I - \Delta t H_n C_n).$$

Denote $M_n := C_n^{1/2} H_n C_n^{1/2}$ and its eigenvalues $\lambda_{1,n}, \ldots, \lambda_{N_\theta,n}$. As $\alpha I \preceq -H_n \preceq \beta I$ and $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$, then

$$-\beta\lambda_{\max} \leq \lambda_{i,n} \leq -\alpha\lambda_{\min}, \qquad i = 1, \ldots, N_\theta,$$

hence the eigenvalues of $C_{n+1}^{-1}C_n$ are bounded by

$$\frac{1+\Delta t\alpha\lambda_{\min}}{1+\Delta t} \leq r_{i,n} = \frac{1-\Delta t\lambda_{i,n}}{1+\Delta t} \leq \frac{1+\Delta t\beta\lambda_{\max}}{1+\Delta t}.$$

We have

$$D^C(C_n, C_{n+1}) = \frac{1}{2}\sum_{i=1}^{N_\theta}\phi(r_{i,n}), \qquad \phi(r) = r - \log r - 1.$$

The Taylor expansion of $\phi(r)$ at $r = 1$ with $\zeta \in \left[\frac{1+\Delta t\alpha\lambda_{\min}}{1+\Delta t}, \frac{1+\Delta t\beta\lambda_{\max}}{1+\Delta t}\right]$ reads

$$\phi(r) = \frac{1}{2}\phi''(\zeta)(r-1)^2 = \frac{1}{2\zeta^2}\cdot(r-1)^2 \geq \frac{1}{2}\left(\frac{1+\Delta t}{1+\Delta t\beta\lambda_{\max}}\right)^2(r-1)^2,$$

hence $D^C$ is bounded below by

$$D^C(C_n, C_{n+1}) \geq \frac{1}{2}\sum_{i=1}^{N_\theta}\frac{1}{2}\left(\frac{1+\Delta t}{1+\Delta t\beta\lambda_{\max}}\right)^2(r_{i,n}-1)^2 \geq \frac{\Delta t^2}{4(1+\Delta t\beta\lambda_{\max})^2}\sum_{i=1}^{N_\theta}(\lambda_{i,n}+1)^2.$$

Since $A_n = C_n^{-1} + H_n = C_n^{-1/2}(I + M_n)C_n^{-1/2}$, then

$$\mathrm{Tr}(A_n C_n A_n) = \mathrm{Tr}(C_n^{-1}(I + M_n)^2) \leq \frac{1}{\lambda_{\min}}\sum_{i=1}^{N_\theta}(\lambda_{i,n}+1)^2.$$

Hence,

$$D^C(C_n, C_{n+1}) \geq \frac{\Delta t^2\lambda_{\min}}{4(1+\Delta t\beta\lambda_{\max})^2}\,\mathrm{Tr}(A_n C_n A_n).$$

These collectively give:

$$\begin{aligned} D(a_n, a_{n+1}) &\geq \frac{\Delta t^2\lambda_{\min}}{2(1+\Delta t)}(1+\Delta t\alpha\lambda_{\min})\|g_n\|^2 + \frac{\Delta t^2\lambda_{\min}}{4(1+\Delta t\beta\lambda_{\max})^2}\,\mathrm{Tr}(A_n C_n A_n) \\ &\geq \kappa(\Delta t)\Delta t^2(\|g_n\|^2 + \mathrm{Tr}(A_n C_n A_n)) = \kappa(\Delta t)\Delta t^2\|\mathrm{grad}^{W_2}\mathcal{E}(a_n)\|_{W_2}^2, \end{aligned} \tag{6}$$

where we denote

$$\kappa(\Delta t) = \lambda_{\min}\min\left\{\frac{1+\Delta t\alpha\lambda_{\min}}{2(1+\Delta t)}, \frac{1}{4(1+\Delta t\beta\lambda_{\max})^2}\right\}.$$

We use the PL-inequality from Wasserstein geometry [10, Appendix D]:

$$\|\mathrm{grad}^{W_2}\mathcal{E}(a_n)\|_{W_2}^2 \geq 2\alpha(\mathcal{E}(a_n) - \mathcal{E}(a_\star)),$$

hence we obtain the PL-type inequality for Bregman divergence:

$$D(a_n, a_{n+1}) \geq 2\alpha \cdot \kappa(\Delta t)\Delta t^2(\mathcal{E}(a_n) - \mathcal{E}(a_\star)).$$

By Lemma 2.9, we have

$$\begin{aligned}\mathcal{E}(a_{n+1}) - \mathcal{E}(a_\star) &\leq \mathcal{E}(a_n) - \mathcal{E}(a_\star) - \frac{1}{\Delta t}D(a_n, a_{n+1}) \\ &\leq (1 - 2\alpha \cdot \kappa(\Delta t)\Delta t)(\mathcal{E}(a_n) - \mathcal{E}(a_\star)),\end{aligned}$$

we therefore obtain the following convergence rate:

$$\mathcal{E}(a_N) - \mathcal{E}(a_\star) \leq (1 - 2\alpha \cdot \kappa(\Delta t)\Delta t)^N(\mathcal{E}(a_0) - \mathcal{E}(a_\star)).$$

□

### 2.3 Convergence of stochastic algorithms

We also consider the convergence theory of the corresponding stochastic algorithm because we may not compute the gradient $\nabla_\theta \log \rho_{\text{post}}(\theta)$ or the Hessian $\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)$ exactly in real-world applications. In the $n$-th step, we first sample $\tilde{\theta} \sim \rho_{a_n}$, and then calculate

$$b_n = \nabla \log \rho_{\text{post}}(\tilde{\theta}), \qquad S_n = \nabla^2 \log \rho_{\text{post}}(\tilde{\theta}),$$

we then compute the stochastic algorithm with Riemannian distance

$$\begin{aligned}m_{n+1} &= m_n + \Delta C_n b_n, \\ C_{n+1} &= e^{\Delta t} C_n^{1/2} \exp(\Delta t C_n^{1/2} S_n C_n^{1/2}) C_n^{1/2},\end{aligned} \tag{7}$$

and the stochastic algorithm with KL divergence

$$\begin{aligned}m_{n+1} &= m_n + \Delta t C_n b_n, \\ C_{n+1} &= (1 + \Delta t)(C_n^{-1} - \Delta t S_n)^{-1}.\end{aligned} \tag{8}$$

We first prove that, under the assumption of the unbiasedness of stochastic estimations, the variances are bounded in both (7) and (8).

**Assumption 2.11.** Suppose $\mathcal{F}_n$ to be a filtration of a discrete algorithm such that $\mathcal{F}_n := \sigma(a_0, \ldots, a_n)$, we assume that the stochastic estimates in (7) and (8) are unbiased, i.e.,

$$\mathbb{E}[b_n \mid \mathcal{F}_n] = \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)], \qquad \mathbb{E}[S_n \mid \mathcal{F}_n] = \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)].$$

**Lemma 2.12.** *Under Assumption 2.1 and Assumption 2.11, for stochastic algorithms (7) and (8), the variances of stochastic estimates are bounded above with*

$$\begin{aligned}\mathbb{E}[\|b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)]\|^2 \mid \mathcal{F}_n] &\leq \beta^2 \lambda_{\max} N_\theta, \\ \mathbb{E}[\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)]\|_F^2 \mid \mathcal{F}_n] &\leq N_\theta(\beta - \alpha)^2,\end{aligned}$$

*where $C_n \preceq \lambda_{\max} I$ along (7) and (8).*

*Proof.* Since $\alpha I \preceq -\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta) \preceq \beta I$, then the map $\nabla_\theta \log\rho_{\text{post}}(\theta)$ is $\beta$-Lipschitz because $\|\nabla_\theta\nabla_\theta \log\rho_{\text{post}}\|_{\text{op}} \le \beta$, this gives

$$\|\nabla \log\rho_{\text{post}}(x) - \nabla\log\rho_{\text{post}}(y)\| \le \beta\|x-y\|, \qquad \forall x, y.$$

Since $\mathbb{E}[b_n \mid \mathcal{F}_n] = \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)]$, then

$$\begin{aligned}\mathbb{E}[\|b_n - \mathbb{E}_{\rho\sim\rho_{a_n}}[\nabla_\theta\log\rho_{\text{post}}(\theta)]\|^2 \mid \mathcal{F}_n] &= \mathbb{E}[\|\nabla\log\rho_{\text{post}}(\tilde\theta) - \mathbb{E}[\nabla\log\rho_{\text{post}}(\tilde\theta) \mid \mathcal{F}_n]\|^2 \mid \mathcal{F}_n] \\ &\le \mathbb{E}[\|\nabla\log\rho_{\text{post}}(\tilde\theta) - \nabla\log\rho_{\text{post}}(m_n)\|^2 \mid \mathcal{F}_n] \\ &\le \beta\mathbb{E}[\|\tilde\theta - m_n\|^2 \mid \mathcal{F}_n] = \beta^2\,\mathrm{Tr}(C_n) \le \beta^2\lambda_{\max}N_\theta.\end{aligned}$$

For the variance bound of Hessian estimator, the stochastic Hessian satisfies $\alpha I \preceq -S_n \preceq \beta I$ almost surely since the posterior distribution satisfies $\alpha I \preceq -\nabla_\theta\nabla_\theta\log\rho_{\text{post}}(\theta) \preceq \beta I$, hence the operator norm of error is bounded above by $\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta\log\rho_{\text{post}}(\theta)]\|_{\text{op}} \le \beta-\alpha$, this gives

$$\mathbb{E}[\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta\log\rho_{\text{post}}(\theta)]\|_F^2 \mid \mathcal{F}_n] \le N_\theta(\beta-\alpha)^2.$$

$\square$

We proceed to show the convergence rate of (7).

**Theorem 2.13.** *Under Assumption 2.1 and Assumption 2.11, we then assume that the initial covariance is bounded with $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$. We write $\lambda_{\min} = \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}$ and $\lambda_{\max} = \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}$, for stepsize $0 < \Delta t \le \frac{1}{\beta\lambda_{\max}}$ on (7), we have the following convergence guarantee:*

$$\mathbb{E}[\mathcal{E}(a_N) - \mathcal{E}(a_\star)] \le (1-\alpha\lambda_{\min}\Delta t(2-\beta\lambda_{\max}\Delta t))^N(\mathcal{E}(a_0)-\mathcal{E}(a_\star)) + \frac{\beta\Delta t\,\lambda_{\max}^3 N_\theta}{2\alpha\lambda_{\min}(2-\beta\lambda_{\max}\Delta t)}\left(\beta^2 + \frac{(\beta-\alpha)^2}{2}\right).$$

*Proof.* We first show the covariance bound along (7). Since $\alpha I \preceq -\nabla_\theta\nabla_\theta\log\rho_{\text{post}}(\theta) \preceq \beta I$, then the stochastic Hessian satisfies $\alpha I \preceq -S_n \preceq \beta I$ almost surely. By Lemma 2.5, along the update (7), the covariance satisfies $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$, where $\lambda_{\min} = \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}$ and $\lambda_{\max} = \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}$ because the stepsize satisfies $0 < \Delta t < \frac{1}{\beta\lambda_{\max}}$.

By Proposition 2.4, the update map of the deterministic algorithm (2) is

$$\overline{a}_{n+1} = \mathrm{Exp}_{a_n}\left(-\Delta t\,\mathrm{grad}\,\mathcal{E}(a_n)\right),$$

similarly, the update map of the stochastic algorithm (7) is

$$a_{n+1} = \mathrm{Exp}_{a_n}\left(-\Delta t\,\widehat{\mathrm{grad}}\,\mathcal{E}(a_n)\right),$$

where

$$\widehat{\mathrm{grad}}\,\mathcal{E}(a_n) = \left(-C_n b_n, -C_n - C_n S_n C_n\right).$$

By the geodesic $L$-smoothness of $\mathcal{E}$, we plug in $a = a_n$ and $\zeta = -\Delta t\,\widehat{\mathrm{grad}}\,\mathcal{E}(a_n)$ into (3) to obtain:

$$\mathcal{E}(a_{n+1}) \le \mathcal{E}(a_n) + \left\langle \mathrm{grad}\mathcal{E}(a_n), -\Delta t\,\widehat{\mathrm{grad}}\,\mathcal{E}(a_n)\right\rangle_{a_n} + \frac{L\Delta t^2}{2}\|\widehat{\mathrm{grad}}\,\mathcal{E}(a_n)\|_{a_n}^2.$$

We set

$$\xi_n := \widehat{\mathrm{grad}}\,\mathcal{E}(a_n) - \mathrm{grad}\,\mathcal{E}(a_n) = \left(-C_n(b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\log\rho_{\text{post}}(\theta)]), -C_n(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta\log\rho_{\text{post}}(\theta)])C_n\right),$$

hence we obtain

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \Delta t \|\operatorname{grad}\mathcal{E}(a_n)\|_{a_n}^2 - \Delta t \langle \operatorname{grad}\mathcal{E}(a_n), \xi_n \rangle_{a_n} + \frac{L\Delta t^2}{2} \|\operatorname{grad}\mathcal{E}(a_n) + \xi_n\|_{a_n}^2.$$

We take conditional expectation to obtain

$$\mathbb{E}[\langle \operatorname{grad}\mathcal{E}(a_n), \xi_n \rangle_{a_n} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\operatorname{grad}\mathcal{E}(a_n) + \xi_n\|_{a_n}^2 \mid \mathcal{F}_n] = \|\operatorname{grad}\mathcal{E}(a_n)\|_{a_n}^2 + \mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n],$$

since $\mathbb{E}[\xi_n \mid \mathcal{F}_n] = 0$ by unbiasedness, hence

$$\mathbb{E}[\mathcal{E}(a_{n+1}) \mid \mathcal{F}_n] \leq \mathcal{E}(a_n) - \Delta t \left(1 - \frac{L\Delta t}{2}\right) \|\operatorname{grad}\mathcal{E}(a_n)\|_{a_n}^2 + \frac{L\Delta t^2}{2} \mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n].$$

We then upper bound $\mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n]$, since the Riemannian metric is given by

$$\|(u, U)\|_{a_n}^2 = u^\top C_n^{-1} u + \frac{1}{2} \operatorname{Tr}(C_n^{-1} U C_n^{-1} U),$$

then for the mean part

$$\begin{aligned}
&(C_n(b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)]))^\top C_n^{-1} (C_n(b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)])) \\
&= (b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)])^\top C_n (b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)]) \leq \lambda_{\max} \|b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log \rho_{\text{post}}(\theta)]\|^2,
\end{aligned}$$

while for the covariance part

$$\begin{aligned}
&\frac{1}{2} \operatorname{Tr}\left(C_n^{-1} \left(C_n(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)])C_n\right) C_n^{-1} \left(C_n(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)])C_n\right)\right) \\
&= \frac{1}{2} \operatorname{Tr}((S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)])C_n (S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)])C_n) \\
&\leq \frac{\lambda_{\max}^2}{2} \|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\text{post}}(\theta)]\|_F^2,
\end{aligned}$$

by the variance bounds in Lemma 2.12, we have

$$\mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n] \leq \lambda_{\max}^2 \beta^2 N_\theta + \frac{\lambda_{\max}^2}{2} (\beta - \alpha)^2 N_\theta.$$

In addition, we connect the gradient norm in Riemannian geometry to the gradient norm in Wasserstein geometry (Theorem 2.3),

$$\|\operatorname{grad}\mathcal{E}(a_n)\|_{a_n}^2 \geq \lambda_{\min} \|\operatorname{grad}^{W_2}\mathcal{E}(a_n)\|_{W_2}^2 \geq 2\alpha\lambda_{\min}(\mathcal{E}(a_n) - \mathcal{E}(a_\star)),$$

then we obtain following one-step inequality:

$$\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(a_\star) \mid \mathcal{F}_n] \leq (1 - \alpha\lambda_{\min}\Delta t\,(2 - L\Delta t))\,(\mathcal{E}(a_n) - \mathcal{E}(a_\star)) + \frac{L\Delta t^2}{2} \left(\lambda_{\max}^2 \beta^2 N_\theta + \frac{\lambda_{\max}^2}{2} (\beta - \alpha)^2 N_\theta\right).$$

We therefore obtain the convergence rate

$$\mathbb{E}[\mathcal{E}(a_N) - \mathcal{E}(a_\star)] \leq (1 - \alpha\lambda_{\min}\Delta t\,(2 - L\Delta t))^N\,(\mathcal{E}(a_0) - \mathcal{E}(a_\star)) + \frac{L\Delta t\,\lambda_{\max}^2 N_\theta}{2\alpha\lambda_{\min}(2 - L\Delta t)} \left(\beta^2 + \frac{(\beta - \alpha)^2}{2}\right).$$

$\square$

We then to show the convergence guarantee of (8).

**Theorem 2.14.** *Under Assumption 2.1 and Assumption 2.11 with initial covariance satisfies* $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$*, we define* $\lambda_{\min} = \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}$ *and* $\lambda_{\max} = \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}$ *and consider the stochastic algorithm* (8) *with stepsize* $0 < \Delta t \leq \frac{1}{L}$ *where* $L = \beta\lambda_{\max}\max\left\{1, \frac{\lambda_{\max}^3}{2\lambda_{\min}^3}\right\}$*, then we have the following the convergence guarantee*

$$\mathbb{E}[\mathcal{E}(a_N) - \mathcal{E}(a_\star)] \leq (1 - 2\alpha\kappa(\Delta t)\Delta t)^N(\mathcal{E}(a_0) - \mathcal{E}(a_\star)) + \frac{B\Delta t}{2\alpha\kappa(\Delta t)},$$

*where*

$$\begin{aligned}\kappa(\Delta t) &= \lambda_{\min}\min\left\{\frac{1+\Delta t\alpha\lambda_{\min}}{2(1+\Delta t)}, \frac{1}{4(1+\Delta t\beta\lambda_{\max})^2}\right\},\\ B &= \frac{\beta}{2}\lambda_{\max}^3(\beta-\alpha)^2 N_\theta^{3/2} + LN_\theta\left(\frac{\beta^2\lambda_{\max}^3}{2\alpha\lambda_{\min}} + \frac{\lambda_{\max}^5(\beta-\alpha)^2}{4\lambda_{\min}^3}\right).\end{aligned}$$

*Proof.* We first show the covariance bound along (8). Since $\alpha I \preceq -\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta) \preceq \beta I$, then the stochastic Hessian satisfies $\alpha I \preceq -S_n \preceq \beta I$ almost surely. By Lemma 2.7, along the update (8), the covariance satisfies $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$, where $\lambda_{\min} = \min\left\{\lambda_{0,\min}, \frac{1}{\beta}\right\}$ and $\lambda_{\max} = \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}$.

We denote the deterministic update (4) as $\overline{a}_{n+1} = (\overline{m}_{n+1}, \overline{C}_{n+1})$ with

$$\begin{aligned}\overline{m}_{n+1} &= m_n + \Delta t C_n \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)],\\ \overline{C}_{n+1} &= (1+\Delta t)(C_n^{-1} - \Delta t\mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)])^{-1},\end{aligned}$$

and analyze the stochastic errors from both mean and covariance directions. For the mean component,

$$m_{n+1} - \overline{m}_{n+1} = \Delta t C_n(b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)]),$$

hence we compute the conditional mean and variance from Lemma 2.12

$$\begin{aligned}\mathbb{E}[m_{n+1} - \overline{m}_{n+1} \mid \mathcal{F}_n] &= 0,\\ \mathbb{E}[\|m_{n+1} - \overline{m}_{n+1}\|^2 \mid \mathcal{F}_n] &= \Delta t^2\mathbb{E}[(b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)])^\top C_n^2(b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)]) \mid \mathcal{F}_n]\\ &\leq \Delta t^2\lambda_{\max}^2\mathbb{E}[\|b_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta \log\rho_{\text{post}}(\theta)]\|^2 \mid \mathcal{F}_n] \leq \Delta t^2\beta^2\lambda_{\max}^3 N_\theta.\end{aligned}$$

For the covariance component, we first denote $P_{n+1} = C_{n+1}^{-1}$, $\overline{P}_{n+1} = \overline{C}_{n+1}^{-1}$, and hence

$$\begin{aligned}\overline{P}_{n+1} &= \frac{1}{1+\Delta t}(P_n - \Delta t\mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)]),\\ P_{n+1} &= \frac{1}{1+\Delta t}(P_n - \Delta t\mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)] - \Delta t(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)])).\end{aligned}$$

This gives $P_{n+1} - \overline{P}_{n+1} = -\frac{\Delta t}{1+\Delta t}(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)])$, hence

$$\begin{aligned}\mathbb{E}[P_{n+1} - \overline{P}_{n+1} \mid \mathcal{F}_n] &= 0,\\ \mathbb{E}[\|P_{n+1} - \overline{P}_{n+1}\|_F^2 \mid \mathcal{F}_n] &= \left(\frac{\Delta t}{1+\Delta t}\right)^2\mathbb{E}[\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)]\|_F^2 \mid \mathcal{F}_n]\\ &\leq \left(\frac{\Delta t}{1+\Delta t}\right)^2(\beta-\alpha)^2 N_\theta.\end{aligned}$$

Then,

$$C_{n+1} - \overline{C}_{n+1} = -\overline{C}_{n+1}(P_{n+1} - \overline{P}_{n+1})C_{n+1}$$
$$= \frac{\Delta t}{1+\Delta t}\overline{C}_{n+1}(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)])C_{n+1},$$

which gives the following bound in error in variance

$$\mathbb{E}[\|C_{n+1} - \overline{C}_{n+1}\|_F^2 \mid \mathcal{F}_n] \leq \left(\frac{\Delta t}{1+\Delta t}\right)^2 \lambda_{\max}^4 \mathbb{E}[\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)]\|_F^2 \mid \mathcal{F}_n]$$
$$\leq \left(\frac{\Delta t}{1+\Delta t}\right)^2 \lambda_{\max}^4 (\beta-\alpha)^2 N_\theta.$$

We then bound the stochastic error for mean, we first write $C_{n+1} = \overline{C}_{n+1} + (C_{n+1} - \overline{C}_{n+1})$ and hence

$$C_{n+1} - \overline{C}_{n+1} = \frac{\Delta t}{1+\Delta t}\overline{C}_{n+1}(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)])\overline{C}_{n+1}$$
$$+ \frac{\Delta t}{1+\Delta t}\overline{C}_{n+1}(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)])(C_{n+1} - \overline{C}_{n+1}).$$

Since $\mathbb{E}[S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)] \mid \mathcal{F}_n] = 0$ and $\overline{C}_{n+1}$ is $\mathcal{F}_n$-measurable, then the first term vanishes after taking conditional expectation and thus

$$\|\mathbb{E}[C_{n+1} - \overline{C}_{n+1} \mid \mathcal{F}_n]\|_F = \left\|\frac{\Delta t}{1+\Delta t}\overline{C}_{n+1}\mathbb{E}[(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)])(C_{n+1} - \overline{C}_{n+1}) \mid \mathcal{F}_n]\right\|_F$$
$$\leq \frac{\Delta t}{1+\Delta t}\lambda_{\max}\mathbb{E}[\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)]\|_F \|C_{n+1} - \overline{C}_{n+1}\|_F \mid \mathcal{F}_n]$$
$$\leq \left(\frac{\Delta t}{1+\Delta t}\right)^2 \lambda_{\max}^3 \mathbb{E}[\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log \rho_{\mathrm{post}}(\theta)]\|_F^2 \mid \mathcal{F}_n]$$
$$\leq \left(\frac{\Delta t}{1+\Delta t}\right)^2 \lambda_{\max}^3 (\beta-\alpha)^2 N_\theta.$$

We proceed to bound the general affine invariant divergence $D$, for the mean part

$$D^m(a_{n+1}, \overline{a}_{n+1}) = \frac{1}{2}(m_{n+1} - \overline{m}_{n+1})^\top \overline{C}_{n+1}^{-1}(m_{n+1} - \overline{m}_{n+1}) \leq \frac{1}{2\lambda_{\min}}\|m_{n+1} - \overline{m}_{n+1}\|^2,$$

hence

$$\mathbb{E}[D^m(a_{n+1}, \overline{a}_{n+1}) \mid \mathcal{F}_n] \leq \frac{\Delta t^2 \beta^2 \lambda_{\max}^3 N_\theta}{2\lambda_{\min}}.$$

For the covariance part, we denote $M = \overline{C}_{n+1}^{-1/2} C_{n+1} \overline{C}_{n+1}^{-1/2}$ with eigenvalues $r_1, \ldots, r_{N_\theta}$ where $\frac{\lambda_{\min}}{\lambda_{\max}} \leq r_i \leq \frac{\lambda_{\max}}{\lambda_{\min}}$, then

$$D^C(a_{n+1}, \overline{a}_{n+1}) = \frac{1}{2}\operatorname{Tr}(M - \log M - I) = \frac{1}{2}\sum_{i=1}^{N_\theta}(r_i - \log r_i - 1)$$
$$\leq \frac{\lambda_{\max}^2}{4\lambda_{\min}^2}\sum_{i=1}^{N_\theta}(r_i - 1)^2 = \frac{\lambda_{\max}^2}{4\lambda_{\min}^2}\|M - I\|_F^2,$$

where the inequality comes from the Taylor expansion of $\phi(r) = r - \log r - 1$ at $r = 1$. Furthermore,

$$
\begin{aligned}
\|M - I\|_F &= \left\|\overline{C}_{n+1}^{-1/2}(C_{n+1} - \overline{C}_{n+1})\overline{C}_{n+1}^{-1/2}\right\|_F \\
&= \frac{\Delta t}{1+\Delta t}\left\|\overline{C}_{n+1}^{1/2}(S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)])C_{n+1}\overline{C}_{n+1}^{-1/2}\right\|_F \\
&\leq \frac{\Delta t}{1+\Delta t}\frac{\lambda_{\max}^{3/2}}{\lambda_{\min}^{1/2}}\|S_n - \mathbb{E}_{\theta\sim\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\text{post}}(\theta)]\|_F,
\end{aligned}
$$

we then take conditional expectation and use Lemma 2.12 to obtain

$$
\mathbb{E}[D^C(a_{n+1}, \overline{a}_{n+1}) \mid \mathcal{F}_n] \leq \left(\frac{\Delta t}{1+\Delta t}\right)^2 \frac{\lambda_{\max}^5}{4\lambda_{\min}^3}(\beta-\alpha)^2 N_\theta.
$$

By $L$-smoothness of $\mathcal{E}_2$ (Proposition 2.8), we have

$$
\mathcal{E}_2(a_{n+1}) \leq \mathcal{E}_2(\overline{a}_{n+1}) + \langle \nabla\mathcal{E}_2(\overline{a}_{n+1}), a_{n+1} - \overline{a}_{n+1}\rangle + LD(a_{n+1}, \overline{a}_{n+1}),
$$

this gives

$$
\begin{aligned}
&\mathbb{E}[\mathcal{E}_2(a_{n+1}) - \mathcal{E}_2(\overline{a}_{n+1}) \mid \mathcal{F}_n] \\
&\leq \mathbb{E}[\langle \nabla_m\mathcal{E}_2(\overline{a}_{n+1}), m_{n+1} - \overline{m}_{n+1}\rangle \mid \mathcal{F}_n] + \mathbb{E}[\langle \nabla_C\mathcal{E}_2(\overline{a}_{n+1}), C_{n+1} - \overline{C}_{n+1}\rangle \mid \mathcal{F}_n] + L\mathbb{E}[D(a_{n+1}, \overline{a}_{n+1}) \mid \mathcal{F}_n] \\
&\leq 0 + \|\nabla_C\mathcal{E}_2(\overline{a}_{n+1})\|_F \cdot \|\mathbb{E}[C_{n+1} - \overline{C}_{n+1} \mid \mathcal{F}_n]\|_F + L\mathbb{E}[D(a_{n+1}, \overline{a}_{n+1}) \mid \mathcal{F}_n] \\
&\leq \frac{\beta}{2}\sqrt{N_\theta}\cdot\left(\frac{\Delta t}{1+\Delta t}\right)^2 \lambda_{\max}^3(\beta-\alpha)^2 N_\theta + LN_\theta\left(\frac{\Delta t^2\beta^2\lambda_{\max}^3}{2\lambda_{\min}} + \left(\frac{\Delta t}{1+\Delta t}\right)^2\frac{\lambda_{\max}^5(\beta-\alpha)^2}{4\lambda_{\min}^3}\right) \leq B\Delta t^2.
\end{aligned}
$$

where we write

$$
B = \frac{\beta}{2}\lambda_{\max}^3(\beta-\alpha)^2 N_\theta^{3/2} + LN_\theta\left(\frac{\beta^2\lambda_{\max}^3}{2\lambda_{\min}} + \frac{\lambda_{\max}^5(\beta-\alpha)^2}{4\lambda_{\min}^3}\right)
$$

and $L = \beta\lambda_{\max}\max\left\{1, \frac{\lambda_{\max}^3}{2\lambda_{\min}^3}\right\}$ from Proposition 2.8.

For $\mathcal{E}_1$, since the function $P \mapsto \log\det P$ is concave, then by Jensen's inequality and $\mathbb{E}[P_{n+1} \mid \mathcal{F}_n] = \overline{P}_{n+1}$, we have

$$
\mathbb{E}[\log\det P_{n+1} \mid \mathcal{F}_n] \leq \log\det\mathbb{E}[P_{n+1} \mid \mathcal{F}_n] = \log\det\overline{P}_{n+1},
$$

this gives

$$
\mathbb{E}[\mathcal{E}_1(a_{n+1}) - \mathcal{E}_1(\overline{a}_{n+1}) \mid \mathcal{F}_n] \leq 0.
$$

Since $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$, then $\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(\overline{a}_{n+1}) \mid \mathcal{F}_n] \leq B\Delta t^2$, we then decompose $\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(a_\star) \mid \mathcal{F}_n]$ to obtain the one-step update from Theorem 2.10:

$$
\begin{aligned}
\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(a_\star) \mid \mathcal{F}_n] &= \mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(\overline{a}_{n+1}) \mid \mathcal{F}_n] + \mathcal{E}(\overline{a}_{n+1}) - \mathcal{E}(a_\star) \\
&\leq B\Delta t^2 + (1 - 2\alpha\cdot\kappa(\Delta t)\Delta t)(\mathcal{E}(a_n) - \mathcal{E}(a_\star)),
\end{aligned}
$$

by iterating on $n$, we obtain the following convergence guarantee:

$$
\mathbb{E}[\mathcal{E}(a_N) - \mathcal{E}(a_\star)] \leq (1 - 2\alpha\kappa(\Delta t)\Delta t)^N(\mathcal{E}(a_0) - \mathcal{E}(a_\star)) + \frac{B\Delta t}{2\alpha\kappa(\Delta t)}.
$$

$\square$

## 3 Local convergence near equilibrium

We proceed to prove the local convergence rate of Gaussian natural gradient flow (1) near equilibrium under Assumption 2.1, where we denote $\kappa = \frac{\beta}{\alpha}$. Based on the evidence from numerical experiments, we expect that the local convergence rate Gaussian natural gradient flow does not rely on initialization as in Theorem 2.3.

We work on the equilibrium-whitened coordinates without loss of generality

$$z = C_\star^{-1/2}(\theta - m_\star) \tag{9}$$

such that the optimality is $a_\star = \mathcal{N}(0, I_{N_\theta})$ and $z \sim \mathcal{N}(0, I_{N_\theta})$, in which case we have $\alpha \leq 1 \leq \beta$. This gives the stationary conditions $\mathbb{E}_{\mathcal{N}(0,I_{N_\theta})}[\nabla V(z)] = 0$ and $\mathbb{E}_{\mathcal{N}(0,I_{N_\theta})}[\nabla^2 V(z)] = I_{N_\theta}$. Hence for any $a = (m, C) \in \mathcal{A}$, we can parameterize $(m, C) = (u, I + X)$ with $u \in \mathbb{R}^{N_\theta}$ and $X \in \mathbb{S}(N_\theta)$.

**Definition 3.1.** For $u \in \mathbb{R}^{N_\theta}$ and $X \in \mathbb{S}(N_\theta)$, define the operators,

$$\begin{aligned}
T[X]_k &:= \mathbb{E}[\mathrm{Tr}(X\nabla^2 V(z))z_k] = \sum_{i,j} \mathbb{E}[\partial^3_{ijk} V(z)] X_{ij}, \\
T^*[u]_{ij} &:= \mathbb{E}[(u^\top z)\nabla^2 V_{ij}(z)], \\
H[X]_{ij} &:= \mathbb{E}[\nabla^2 V_{ij}(z)(z^\top X z - \mathrm{Tr}\, X)] = \sum_{k,l} \mathbb{E}[\partial^4_{ijkl} V(z)] X_{kl},
\end{aligned}$$

where we have applied Stein's identity in $\mathcal{T}$ and $\mathcal{H}$. For the diagonal mode, we define

$$\widetilde{T}_{ki} := \mathbb{E}[\nabla^2 V_{ii}(z) z_k], \qquad G_{ij} := \mathbb{E}[\nabla^2 V_{ii}(z) z_j^2].$$

We first show the tail-cut bound of $G_{ij}$.

**Lemma 3.2** (Tail-cut bound)**.** *Under Assumption 2.1 and (9), for any $i, j$, we have*

$$\alpha \leq G_{ij} \leq 4 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa).$$

*Proof.* Since $\alpha I \preceq \nabla^2 V \preceq \beta I$, then $\alpha \leq \nabla^2 V_{ii}(z) \leq \beta$, hence $G_{ij} = \mathbb{E}[\nabla^2 V_{ii}(z) z_j^2] \geq \alpha \mathbb{E}[z_j^2] = \alpha$ since $z_j \sim \mathcal{N}(0, 1)$. This proves the lower bound. For the upper bound, if $\kappa \leq e^{1/2}$, then $G_{ij} \leq \beta \leq \kappa \leq e^{1/2} < 4 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa)$, so we consider the case where $\kappa > e^{1/2}$ and choose $t_0 = \sqrt{2 \log \kappa} > 1$. We split the expectation into the bulk region and the tail region:

$$G_{ij} = \mathbb{E}[\nabla^2 V_{ii}(z) z_j^2] = \mathbb{E}[\nabla^2 V_{ii}(z) z_j^2 \mathbf{1}_{|z_j| < t_0}] + \mathbb{E}[\nabla^2 V_{ii}(z) z_j^2 \mathbf{1}_{|z_j| \geq t_0}].$$

For the bulk part,

$$\mathbb{E}[\nabla^2 V_{ii}(z) z_j^2 \mathbf{1}_{|z_j| < t_0}] \leq t_0^2 \mathbb{E}[\nabla^2 V_{ii}(z)] = 2 \log \kappa.$$

For the tail part, we define $\varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$, then

$$\mathbb{E}[\nabla^2 V_{ii}(z) z_j^2 \mathbf{1}_{|z_j| \geq t_0}] \leq \kappa \mathbb{E}[z_j^2 \mathbf{1}_{|z_j| \geq t_0}] = 2\kappa \int_{t_0}^{\infty} t^2 \varphi(t) \mathrm{d}t,$$

using integration by parts and Mills ratio,

$$\int_{t_0}^{\infty} t^2 \varphi(t) \mathrm{d}t = t_0 \varphi(t_0) + \int_{t_0}^{\infty} \varphi(t) \mathrm{d}t \leq t_0 \varphi(t_0) + \frac{\varphi(t_0)}{t_0} \leq 2 t_0 \varphi(t_0) = 2\sqrt{\frac{\log \kappa}{\pi}} \kappa^{-1}.$$

Hence,

$$G_{ij} \le 2\log\kappa + \frac{4}{\sqrt{\pi}}\sqrt{\log\kappa} \le 4 + \frac{4}{\sqrt{\pi}}(1+\log\kappa).$$

□

We then prove an identity which characterizes the "perturbation" near equilibrium.

**Lemma 3.3.** *Under Assumption 2.1 and (9), for any $u \in \mathbb{R}^{N_\theta}$, $X \in \mathbb{S}(N_\theta)$, and $\eta(z) := u + Xz$,*

$$\mathbb{E}[\eta(z)^\top \nabla^2 V(z)\eta(z)] = \|u\|^2 + 2u^\top T[X] + \|X\|_F^2 + \mathrm{Tr}(XH[X]).$$

*Specifically when $X = \mathrm{diag}\,\lambda$,*

$$\mathbb{E}[\eta(z)^\top \nabla^2 V(z)\eta(z)] = \|u\|^2 + 2u^\top \widetilde{T}\lambda + \|\lambda\|^2 + \lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda.$$

*Proof.* We expand

$$\eta^\top \nabla^2 V \eta = u^\top \nabla^2 V u + 2u^\top \nabla^2 V(Xz) + (Xz)^\top \nabla^2 V(Xz)$$

and analyze the expectation for each term. For the constant term, since $\mathbb{E}[\nabla^2 V] = I$, then $\mathbb{E}[u^\top \nabla^2 V u] = u^\top \mathbb{E}[\nabla^2 V]u = \|u\|^2$. For the cross term, by Stein's Lemma and the definition of $T$,

$$\mathbb{E}[u^\top \nabla^2 V(Xz)] = \sum_{a,b,c} u_a X_{bc}\mathbb{E}[\nabla^2 V_{ab}(z)z_c] = \sum_{a,b,c} u_a X_{bc}\mathbb{E}[\partial_a\partial_b\partial_c V(z)] = u^\top T[X].$$

For the quadratic term, we apply Stein's Lemma twice to obtain

$$\begin{aligned}\mathbb{E}[(Xz)^\top \nabla^2 V(Xz)] &= \sum_{a,b,c,d} X_{ac}X_{bd}\mathbb{E}[\nabla^2 V_{ab}(z)z_c z_d] \\ &= \sum_{a,b,c,d} X_{ac}X_{bd}(\mathbb{E}[\partial_a\partial_b\partial_c\partial_d V(z)] + \delta_{cd}\delta_{ab}) = \mathrm{Tr}(XH[X]) + \|X\|_F^2.\end{aligned}$$

Adding these three terms up completes the proof. □

Based on Lemma 3.3, we aim to prove the spectral upper bound of $\widetilde{T}^\top \widetilde{T}$.

**Lemma 3.4.** *Under Assumption 2.1 and (9),*

$$\widetilde{T}^\top \widetilde{T} \preceq I + (G - \mathbf{1}\mathbf{1}^\top).$$

*Proof.* We choose $\eta(z) = u + \mathrm{diag}(\lambda)z$ in Lemma 3.3, since $\nabla^2 V(z) \succeq 0$, then

$$0 \le \|u\|^2 + 2u^\top \widetilde{T}\lambda + \|\lambda\|^2 + \lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda$$

for any $u \in \mathbb{R}^{N_\theta}$ and $\lambda \in \mathbb{R}^{N_\theta}$, which is equivalent to

$$\begin{pmatrix} I & \widetilde{T} \\ \widetilde{T} & I + (G - \mathbf{1}\mathbf{1}^\top) \end{pmatrix} \succeq 0.$$

As $I \succeq 0$, then we have $I + (G - \mathbf{1}\mathbf{1}^\top) - \widetilde{T}^\top \widetilde{T} \succeq 0$ by Schur complement criterion, which completes the proof. □

Based on these Lemmas, we prove the following operator bound.

**Lemma 3.5** (Operator bound)**.** *Under Assumption 2.1 and* (9)*,*

$$\lambda_{\max}(G - \mathbf{1}\mathbf{1}^\top) \le \Gamma := \min\left\{\beta - 1, N_\theta\left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right)\right\}.$$

*Proof.* We prove two spectral upper bounds of $G - \mathbf{1}\mathbf{1}^\top$ separately. On the one hand, we take $\eta(z) = \mathrm{diag}(\lambda)z$ as in Lemma 3.3, this gives

$$\mathbb{E}[\eta(z)^\top \nabla^2 V(z)\eta(z)] = \|\lambda\|^2 + \lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda$$

and concurrently

$$\mathbb{E}[\eta(z)^\top \nabla^2 V(z)\eta(z)] \le \beta\mathbb{E}\|\mathrm{diag}(\lambda)z\|^2 = \beta\|\mathrm{diag}\,\lambda\|_F^2 = \beta\|\lambda\|^2.$$

Collectively, $\lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda \le (\beta - 1)\|\lambda\|^2$, hence $\lambda_{\max}(G - \mathbf{1}\mathbf{1}^\top) \le \beta - 1$.

On the other hand, by Lemma 3.2, we obtain

$$|G_{ij} - 1| \le 3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa),$$

this gives

$$\begin{aligned}
\lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda &= \sum_{i,j}\lambda_i\lambda_j(G_{ij} - 1) \le \sum_{i,j}|\lambda_i|\cdot|\lambda_j|\left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right) \\
&\le \left(\sum_i |\lambda_i|\right)^2\left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right) \le \left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right)N_\theta\|\lambda\|^2,
\end{aligned}$$

this gives $\lambda_{\max}(G - \mathbf{1}\mathbf{1}^\top) \le \left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right)N_\theta$. Hence collectively,

$$\lambda_{\max}(G - \mathbf{1}\mathbf{1}^\top) \le \min\left\{\beta - 1, N_\theta\left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right)\right\}.$$

□

We then compute the linearized Jacobian matrix of Gaussian natural gradient flow (1) and bound its largest eigenvalue.

**Proposition 3.6.** *We consider the Gaussian natural gradient flow* (1) *under Assumption 2.1 and* (9)*. For $u = m \in \mathbb{R}^{N_\theta}$ and $X = C - I \in \mathbb{S}(N_\theta)$, the linearized Jacobian matrix at equilibrium is given by*

$$-J_\star(u, X) = \left(u + \frac{1}{2}T[X], X + T^*[u] + \frac{1}{2}H[X]\right).$$

*Moreover, the largest eigenvalue $\lambda_{\star,\max}$ of $J_\star$ satisfies*

$$-\lambda_{\star,\max} \ge \frac{1}{4 + \Gamma} = \frac{1}{4 + \min\left\{\beta - 1, N_\theta\left(3 + \frac{4}{\sqrt{\pi}}(1 + \log\kappa)\right)\right\}},$$

*and the smallest eigenvalue $\lambda_{\star,\min}$ of $J_\star$ satisfies*

$$-\lambda_{\star,\min} \le \beta + \frac{1 - \alpha}{2}.$$

*Proof.* We apply Bonnet's identity and Price's identity, i.e.,

$$\partial_m \mathbb{E}_{\mathcal{N}(m,C)}[f] = \mathbb{E}_{\mathcal{N}(m,C)}[\nabla f], \qquad \partial_C \mathbb{E}_{\mathcal{N}(m,C)}[f] = \frac{1}{2}\mathbb{E}_{\mathcal{N}(m,C)}[\nabla^2 f],$$

by Taylor's expansion near $a_\star$, we obtain

$$\begin{aligned}\mathbb{E}_{\mathcal{N}(u,I+X)}[\nabla V] &= u + \frac{1}{2}T[X] + O(\|(u,X)\|^2),\\ \mathbb{E}_{\mathcal{N}(u,I+X)}[\nabla^2 V] &= I + T^*[u] + \frac{1}{2}H[X] + O(\|(u,X)\|^2).\end{aligned}$$

By substituting $\mathbb{E}_{\mathcal{N}(u,I+X)}[\nabla V]$ and $\mathbb{E}_{\mathcal{N}(u,I+X)}[\nabla^2 V]$ into (1), we obtain the local ODEs

$$\begin{cases}\dot{u} &= -u - \frac{1}{2}T[X] + O(\|(u,X)\|_\star^2),\\ \dot{X} &= -X - T^*[u] - \frac{1}{2}H[X] + O(\|(u,X)\|_\star^2).\end{cases}$$

Therefore the linearized Jacobian matrix is

$$-J_\star(u,X) = \left(u + \frac{1}{2}T[X], X + T^*[u] + \frac{1}{2}H[X]\right).$$

We proceed to show that the operator $J_\star$ is self-adjoint with respect to $\langle\cdot,\cdot\rangle_\star^{\mathrm{FR}}$. For $(u_1,X_1),(u_2,X_2) \in \mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$, we calculate

$$\langle -J_\star(u_1,X_1),(u_2,X_2)\rangle_\star^{\mathrm{FR}} = u_1^\top u_2 + \frac{1}{2}T[X_1]^\top u_2 + \frac{1}{2}\operatorname{Tr}(X_1X_2) + \frac{1}{2}\operatorname{Tr}(T^*[u_1]X_2) + \frac{1}{4}\operatorname{Tr}(H[X_1]X_2),$$

since $u^\top T[X] = \operatorname{Tr}(T^*[u]X)$ and $\operatorname{Tr}(H[X_1]X_2) = \operatorname{Tr}(X_1 H[X_2])$, then the expression is symmetric, which justifies the self-adjointness.

Thus by the spectral theorem, we obtain

$$-\lambda_{\star,\max} = \min_{(u,X)\neq 0} \frac{\langle -J_\star(u,X),(u,X)\rangle_\star^{\mathrm{FR}}}{\langle (u,X),(u,X)\rangle_\star^{\mathrm{FR}}} = \min_{(u,X)\neq 0}\frac{Q(u,X)}{\|u\|^2 + \frac{1}{2}\|X\|_F^2},$$

where $Q(u,X) := \|u\|^2 + u^\top T[X] + \frac{1}{2}\|X\|_F^2 + \frac{1}{4}\operatorname{Tr}(XH[X])$. Since the standard Gaussian measure and the Hessian bounds are invariant under orthogonal changes of basis, we work in orthogonal basis $X = \operatorname{diag}\lambda$, hence

$$Q(u,\operatorname{diag}\lambda) = \|u\|^2 + u^\top \widetilde{T}\lambda + \frac{1}{2}\|\lambda\|^2 + \frac{1}{4}\lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda.$$

We hence only need to show $Q(u,\operatorname{diag}\lambda) \geq \frac{1}{4+\Gamma}(\|u\|^2 + \frac{1}{2}\|\lambda\|^2)$, which is equivalent to

$$(1-\nu)\|u\|^2 + u^\top\widetilde{T}\lambda + \frac{1-\nu}{2}\|\lambda\|^2 + \frac{1}{4}\lambda^\top(G - \mathbf{1}\mathbf{1}^\top)\lambda \geq 0,$$

where we denote $\nu = \frac{1}{4+\Gamma}$ and $0 < \nu \leq \frac{1}{4}$. It suffices to show that

$$\frac{1-\nu}{2}I + \frac{1}{4}(G - \mathbf{1}\mathbf{1}^\top) - \frac{\widetilde{T}^\top\widetilde{T}}{4(1-\nu)} \succeq 0,$$

from Lemma 3.4 and Lemma 3.5, a sufficient condition is

$$\frac{1-\nu}{2} - \frac{1+\nu\Gamma}{4(1-\nu)} \geq 0 \Longleftrightarrow f(\nu) := 1 - \nu(4+\Gamma) + 2\nu^2 \geq 0.$$

The smaller root of $f$ is

$$\nu_1 = \frac{(4+\Gamma) - \sqrt{(4+\Gamma)^2 - 8}}{4} = \frac{2}{4+\Gamma+\sqrt{(4+\Gamma)^2-8}} \geq \frac{1}{4+\Gamma},$$

hence $f(\frac{1}{4+\Gamma}) \geq 0$, which completes the proof of the largest eigenvalue bound.

We proceed to bound the smallest eigenvalue of $J_\star$. From Lemma 3.3 and Assumption 2.1, we have

$$\mathbb{E}[(u+Xz)^\top \nabla^2 V(z)(u+Xz)] = \|u\|^2 + 2u^\top T[X] + \|X\|_F^2 + \operatorname{Tr}(XH[X]) \leq \beta(\|u\|^2 + \|X\|_F^2),$$

also by taking $u = 0$, we obtain

$$\|X\|_F^2 + \operatorname{Tr}(XH[X]) \geq \alpha \|X\|_F^2 \implies \operatorname{Tr}(XH[X]) \geq (\alpha - 1)\|X\|_F^2.$$

We then upper bound $Q(u,X)$ by $\|(u,X)\|_\star^2$:

$$\begin{aligned}
Q(u,X) &= \|u\|^2 + u^\top T[X] + \frac{1}{2}\|X\|_F^2 + \frac{1}{4}\operatorname{Tr}(XH[X]) \\
&= \frac{1}{2}\mathbb{E}[(u+Xz)^\top \nabla^2 V(z)(u+Xz)] + \frac{1}{2}\|u\|^2 - \frac{1}{4}\operatorname{Tr}(XH[X]) \\
&\leq \frac{\beta+1}{2}\|u\|^2 + \frac{2\beta+1-\alpha}{4}\|X\|_F^2 \\
&\leq \max\left\{\frac{\beta+1}{2}, \frac{2\beta+1-\alpha}{2}\right\}\|(u,X)\|_\star^2 \leq \left(\beta + \frac{1-\alpha}{2}\right)\|(u,X)\|_\star^2,
\end{aligned}$$

where the last inequality comes from $\alpha \leq 1 \leq \beta$ following (9). Then by spectral theorem,

$$-\lambda_{\star,\min} \leq \beta + \frac{1-\alpha}{2}.$$

□

We provide local convergence rate for the continuous dynamics (1). Before that, we first provide explicit second-order bound for the natural gradient vector field.

**Lemma 3.7.** *Under Assumption 2.1 and (9), we define the vector field of (1) as*

$$F(m,C) = (F_1(m,C), F_2(m,C)) := (-C\mathbb{E}_{\mathcal{N}(m,C)}[\nabla V], C - C\mathbb{E}_{\mathcal{N}(m,C)}[\nabla^2 V]C),$$

*then on the neighborhood $\mathcal{S}_\star = \left\{(m,C) : \frac{1}{2}I \preceq C \preceq \frac{3}{2}I\right\}$, the second differential of vector field $F$ satisfies*

$$\|D^2F(a)[\zeta,\zeta]\|_\star \leq 135\beta N_\theta^{5/2}\|\zeta\|_\star^2,$$

*where $a = (m,C) \in \mathcal{S}_\star$ and the tangent direction $\zeta = (v,Y) \in \mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$.*

*Proof.* We consider the line $a(s) = (m + sv, C + sY)$, since $\|\zeta\|_\star^2 = \|v\|^2 + \frac{1}{2}\|Y\|_F^2$, then we have $\|v\| \leq \|\zeta\|_\star$ and $\|Y\|_{\text{op}} \leq \|Y\|_F \leq \sqrt{2}\|\zeta\|_\star$. For $\theta \in \mathcal{N}(m,C)$ with $(m,C) \in \mathcal{S}_\star$, we denote $w := C^{-1}(\theta - m) = C^{-1/2}z$ for $z \in \mathcal{N}(0, I_{N_\theta})$. Using the standard Gaussian bounds, we obtain the bounds of moments of $w$,

$$\mathbb{E}\|w\| \leq \sqrt{2}N_\theta, \quad \mathbb{E}\|w\|^2 \leq 2N_\theta, \quad \mathbb{E}\|w\|^3 \leq 2\sqrt{2}N_\theta\sqrt{N_\theta+2}, \quad \mathbb{E}\|w\|^4 \leq 4N_\theta(N_\theta+2).$$

For any smooth function $f$, we apply Bonnet–Price differentiation to obtain

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}_{\mathcal{N}(m+sv,C+sY)}[f] = \langle v, \nabla_m \mathbb{E}[f]\rangle + \langle Y, \nabla_C \mathbb{E}[f]\rangle_F = \mathbb{E}\left[v^\top \nabla f + \frac{1}{2}\operatorname{Tr}(Y\nabla^2 f)\right],$$

which is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[f] = \mathbb{E}[\mathcal{L}_\zeta f], \qquad \mathcal{L}_\zeta = \sum_p v_p \partial_p + \frac{1}{2}\sum_{p,q} Y_{pq}\partial_p\partial_q,$$

for the second derivative, we have

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[f] = \mathbb{E}[\mathcal{L}_\zeta^2 f], \qquad \mathcal{L}_\zeta^2 = \sum_{p,q} v_p v_q \partial_p\partial_q + \sum_{p,q,r} v_p Y_{qr}\partial_p\partial_q\partial_r + \frac{1}{4}\sum_{p,q,r,s} Y_{pq}Y_{rs}\partial_p\partial_q\partial_r\partial_s.$$

We thus compute the first and second derivative of $\mathbb{E}[\nabla V]$ and $\mathbb{E}[\nabla^2 V]$ using Gaussian integration by parts:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[\nabla V] &= \mathbb{E}\left[\nabla^2 V\left(v + \frac{1}{2}Yw\right)\right],\\
\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[\nabla^2 V] &= \mathbb{E}\left[\nabla^2 V\left(v^\top w + \frac{1}{2}w^\top Y w - \frac{1}{2}\operatorname{Tr}(YC^{-1})\right)\right],\\
\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[\nabla V] &= \mathbb{E}\left[\nabla^2 V\left((v^\top w)v + (v^\top w)Yw - YC^{-1}v + \frac{1}{4}(w^\top Y w - \operatorname{Tr}(YC^{-1}))Yw - \frac{1}{2}YC^{-1}Yw\right)\right],\\
\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[\nabla^2 V] &= \mathbb{E}\left[\nabla^2 V\left((v^\top w)^2 - v^\top C^{-1}v + (v^\top w)(w^\top Y w) - 2v^\top C^{-1}Yw - (v^\top w)\operatorname{Tr}(YC^{-1}) + \frac{1}{4}\Pi\right)\right],
\end{aligned}$$

where $\Pi = (w^\top Y w)^2 - 2\operatorname{Tr}(YC^{-1})(w^\top Y w) - 4w^\top YC^{-1}Yw + \operatorname{Tr}(YC^{-1})^2 + 2\operatorname{Tr}(C^{-1}YC^{-1}Y)$. Using $\|\nabla^2 V\|_{\mathrm{op}} \le \beta$, $\|\nabla^2 V\|_F \le \sqrt{N_\theta}\beta$, we bound each term:

$$\begin{aligned}
\left\|\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[\nabla V]\right\| &\le \beta(1+\sqrt{N_\theta})\|\zeta\|_\star,\\
\left\|\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[\nabla^2 V]\right\|_F &\le \sqrt{2}\beta(N_\theta^{3/2} + 2N_\theta)\|\zeta\|_\star,\\
\left\|\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[\nabla V]\right\| &\le \sqrt{2}\beta\left(N_\theta\sqrt{N_\theta+2} + 3N_\theta + 3\sqrt{N_\theta} + 2\right)\|\zeta\|_\star^2,\\
\left\|\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[\nabla^2 V]\right\|_F &\le \sqrt{N_\theta}\beta\left(2N_\theta^2 + 4N_\theta\sqrt{N_\theta+2} + 4N_\theta^{3/2} + 18N_\theta + 8\sqrt{N_\theta} + 8\right)\|\zeta\|_\star^2.
\end{aligned}$$

Differentiating $F_1$ and $F_2$ twice along $a(s)$ and substituting the bounds above gives:

$$\begin{aligned}
\|D^2 F_1[\zeta,\zeta]\| &\le \frac{3}{2}\left\|\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[\nabla V]\right\| + 2\sqrt{2}\|\zeta\|_\star\left\|\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[\nabla V]\right\|,\\
\|D^2 F_2[\zeta,\zeta]\|_F &\le \frac{9}{4}\left\|\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathbb{E}[\nabla^2 V]\right\|_F + 6\sqrt{2}\|\zeta\|_\star\left\|\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[\nabla^2 V]\right\|_F + 4\beta\|\zeta\|_\star^2.
\end{aligned}$$

By the definition of the Fisher–Rao norm, $\|D^2F[\zeta,\zeta]\|_\star \le \|D^2F_1[\zeta,\zeta]\| + \frac{1}{\sqrt{2}}\|D^2F_2[\zeta,\zeta]\|_F$. Summing the bounds and evaluating the coefficients using $\sqrt{N_\theta+2} \le \sqrt{3}\sqrt{N_\theta}$ ($N_\theta \ge 1$) yields:

$$\|D^2F[\zeta,\zeta]\|_\star \le 135\beta N_\theta^{5/2}\|\zeta\|_\star^2.$$

□

**Theorem 3.8.** *Under Assumption 2.1 and (9), we denote*

$$\delta := \min\left\{\frac{1}{2\sqrt{2}}, \frac{1}{135\beta N_\theta^{5/2}(4+\Gamma)}\right\}, \qquad \Gamma = \min\left\{\beta - 1, N_\theta\left(3 + \frac{4}{\sqrt{\pi}}(1+\log\kappa)\right)\right\}.$$

*If the initialization satisfies $\|a_0 - a_\star\|_\star \le \delta$, then the local convergence rate of the Gaussian natural gradient flow (1) reads*

$$\|a_t - a_\star\|_\star^2 \le \exp\left(-\frac{t}{4+\Gamma}\right)\|a_0 - a_\star\|_\star^2, \qquad t \ge 0.$$

*Proof.* We denote $\xi_t = a_t - a_\star$. Since the vector field of (1) satisfies $F(a_\star) = 0$ with $DF(a_\star) = J_\star$, we apply Taylor's theorem to obtain

$$F(a_\star + \xi) = J_\star\xi + R(\xi), \qquad R(\xi) = \int_0^1 (1-s)D^2F(a_\star + s\xi)[\xi,\xi]\mathrm{d}s.$$

By Lemma 3.7, since $\|\xi\|_\star \le \delta \le \frac{1}{2\sqrt{2}}$, then the trajectory remains in the convex neighborhood $\mathcal{S}_\star$, hence the remainder satisfies $\|R(\xi)\|_\star \le \frac{135}{2}\beta N_\theta^{5/2}\|\xi\|_\star^2$.

We proceed to compute and bound the derivative of the squared norm

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\|\xi_t\|_\star^2 &= 2\langle \xi_t, J_\star\xi_t\rangle_\star + 2\langle \xi_t, R(\xi_t)\rangle_\star \\ &\le -\frac{2}{4+\Gamma}\|\xi_t\|_\star^2 + 135\beta N_\theta^{5/2}\|\xi_t\|_\star^3 \le -\frac{1}{4+\Gamma}\|\xi_t\|_\star^2,\end{aligned}$$

where the first inequality comes from Proposition 3.6 and Lemma 3.7 while the second inequality comes from $\|\xi_t\|_\star \le \delta \le \frac{1}{135\beta N_\theta^{5/2}(4+\Gamma)}$. This completes the proof by Gronwall's inequality. □

We then show the local convergence rate of the discrete algorithm with Riemannian distance (2).

**Theorem 3.9.** *Under Assumption 2.1 and (9), we define*

$$\delta_{\mathrm{R}}(\Delta t) := \min\left\{\frac{1}{2\sqrt{2}}, \frac{1}{(4+\Gamma)\cdot 148\beta^2 N_\theta^{5/2}}\right\}, \quad \Gamma = \min\left\{\beta - 1, N_\theta\left(3 + \frac{4}{\sqrt{\pi}}(1+\log\kappa)\right)\right\}.$$

*Consider the discrete update with Riemannian distance (2) with stepsize $0 < \Delta t \le \left(\beta + \frac{1-\alpha}{2}\right)^{-1}$, if the initialization satisfies $\|a_0 - a_\star\|_\star \le \delta_{\mathrm{R}}(\Delta t)$, then the local convergence rate is*

$$\|a_N - a_\star\|_\star \le \left(1 - \frac{\Delta t}{2(4+\Gamma)}\right)^N \|a_0 - a_\star\|_\star.$$

*Proof.* We denote $\xi_n := a_n - a_\star$. By Proposition 2.4, for the discrete update (2), we have

$$a_{n+1} = \mathrm{Exp}_{a_n}(-\Delta t\,\mathrm{grad}\,\mathcal{E}(a_n)) = \mathrm{Exp}_{a_n}(\Delta t F(a_n)).$$

The update of error admits the following expansion

$$\xi_{n+1} = (I + \Delta t J_\star)\xi_n + R_{\mathrm{R},\Delta t}(\xi_n).$$

By Proposition 3.6, the linear part satisfies

$$\|(I+\Delta t J_\star)\xi\|_\star \le \left(1-\frac{\Delta t}{4+\Gamma}\right)\|\xi_n\|_\star$$

since the stepsize satisfies $0<\Delta t \le \left(\beta+\frac{1-\alpha}{2}\right)^{-1}$.

We proceed to look into the nonlinear term $R_{\mathrm{R},\Delta t}$. From Lemma 3.7, the vector field part contributes

$$\Delta t\left\|\int_0^1 (1-s)D^2F(a_\star+s\xi_n)[\xi_n,\xi_n]\mathrm{d}s\right\|_\star \le \frac{\Delta t}{2}135\beta N_\theta^{5/2}\|\xi_n\|_\star.$$

The remaining part comes from the exponential covariance increment in the covariance component. We denote

$$P(a)=I+C^{1/2}\mathbb{E}_{\rho_a}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}]C^{1/2}$$

and bound the first variation of $P$ at $a_\star$ for $\xi=(u,X)$:

$$DP(a_\star)[\xi]=-\left(X+T^*[u]+\frac{1}{2}H[X]\right)\Longrightarrow \|DP(a_\star)[\xi]\|_F\le\left(\sqrt{2}+\frac{\beta-\alpha}{2}+\beta N_\theta\right)\|\xi\|_\star.$$

We expand the covariance update $C_{n+1}=C_n^{1/2}\exp(\Delta t P(a_n))C_n^{1/2}$, since

$$\exp(\Delta t P)=I+\Delta t P+\frac{\Delta t^2}{2}P^2+O(\Delta t^3\|P\|^3),$$

we obtain

$$\begin{aligned}\|R_{\mathrm{R},\Delta t}(\xi_n)\|_\star &\le \frac{\Delta t}{2}135\beta N_\theta^{5/2}\|\xi_n\|_\star^2+\frac{\Delta t}{2}\left(2\left(\sqrt{2}+\frac{\beta-\alpha}{2}+\beta N_\theta\right)+\frac{1}{\sqrt{2}}\left(\sqrt{2}+\frac{\beta-\alpha}{2}+\beta N_\theta\right)^2\Delta t\right)\|\xi_n\|_\star^2\\ &\le \frac{\Delta t}{2}\left(135\beta N_\theta^{5/2}+6\beta N_\theta+\frac{9\beta^2N_\theta^2}{\sqrt{2}}\right)\|\xi_n\|_\star^2\le\frac{\Delta t}{2}\cdot 148\beta^2N_\theta^{5/2}\|\xi_n\|_\star^2,\end{aligned}$$

where the continuous-field component from the first inequality comes from $\|\xi_n\|_\star\le\delta_{\mathrm{R}}(\Delta t)\le\frac{1}{2\sqrt{2}}$, which ensures $a_n\in\mathcal{S}_\star$.

Therefore

$$\begin{aligned}\|\xi_{n+1}\|_\star &\le \|(I+\Delta t J_\star)\xi_n\|_\star+\|R_{\mathrm{R},\Delta t}(\xi)\|_\star\\ &\le\left(1-\frac{\Delta t}{4+\Gamma}\right)\|\xi_n\|_\star+\frac{\Delta t}{2}148\beta^2N_\theta^{5/2}\|\xi_n\|_\star^2\\ &\le\left(1-\frac{\Delta t}{2(4+\Gamma)}\right)\|\xi_n\|_\star,\end{aligned}$$

where the last inequality comes from $\|\xi_n\|_\star\le\delta_{\mathrm{R}}(\Delta t)\le\frac{1}{(4+\Gamma)148\beta^2N_\theta^{5/2}}$. This completes the proof by iterating on $n$. □

We proceed to proceed to prove the local convergence rate of discretization scheme with KL divergence (4).

**Theorem 3.10.** *Under Assumption 2.1 and (9), we define*

$$\delta_{\mathrm{KL}}(\Delta t) := \min\left\{\frac{1}{2\sqrt{2}}, \frac{1}{(4+2\Delta t+\Gamma)\cdot 216\beta^2 N_\theta^{5/2}}\right\}, \quad \Gamma = \min\left\{\beta - 1, N_\theta\left(3+\frac{4}{\sqrt{\pi}}(1+\log\kappa)\right)\right\}.$$

*Consider the discrete update with KL divergence (4) with the following local norm*

$$\|(u,X)\|_{\star,\Delta t}^2 = \|u\|^2 + \frac{1+\Delta t}{2}\|X\|_F^2, \qquad (u,X)\in\mathbb{R}^{N_\theta}\times\mathbb{S}(N_\theta),$$

*if the stepsize satisfies $0 < \Delta t \leq \left(\beta + \frac{1-\alpha}{2}\right)^{-1}$ and the initialization satisfies $\|a_0 - a_\star\|_{\star,\Delta t} \leq \delta_{\mathrm{KL}}(\Delta t)$, then the local convergence guarantee reads*

$$\|a_N - a_\star\|_{\star,\Delta t} \leq \left(1 - \frac{\Delta t}{2(4+2\Delta t+\Gamma)}\right)^N \|a_0 - a_\star\|_{\star,\Delta t}.$$

*Proof.* We write $\xi_n := a_n - a_\star = (u_n, X_n)$, and characterize (4) into linear part and non-linear part

$$\xi_{n+1} = M_{\Delta t,\star}^{\mathrm{KL}}(\xi_n) + R_{\Delta t}^{\mathrm{KL}}(\xi_n).$$

We first calculate the linearized map using the similar technique in Proposition 3.6:

$$M_{\Delta t,\star}^{\mathrm{KL}}(u,X) = \left((1-\Delta t)u - \frac{\Delta t}{2}T[X],\ \frac{1}{1+\Delta t}\left(X - \Delta t T^*[u] - \frac{\Delta t}{2}H[X]\right)\right),$$

which is self-adjoint with respect to the weighted inner product at equilibrium because

$$\langle M_{\Delta t,\star}^{\mathrm{KL}}(u_1,X_1),(u_2,X_2)\rangle_{\star,\Delta t} = \langle (u_1,X_1), M_{\Delta t,\star}^{\mathrm{KL}}(u_2,X_2)\rangle_{\star,\Delta t},$$

where the inner product at equilibrium is defined by

$$\langle (u_1,X_1),(u_2,X_2)\rangle_{\star,\Delta t} := u_1^\top u_2 + \frac{1+\Delta t}{2}\,\mathrm{Tr}(X_1X_2).$$

We observe that

$$\|(u,X)\|_{\star,\Delta t}^2 - \langle M_{\Delta t,\star}^{\mathrm{KL}}(u,X),(u,X)\rangle_{\star,\Delta t} = \Delta t Q(u,X),$$

hence the largest eigenvalue $\rho_\star^{\mathrm{KL}}$ of the linearized Jacobian matrix $M_{\Delta t,\star}^{\mathrm{KL}}$ satisfies

$$1 - \rho_\star^{\mathrm{KL}} = \Delta t \min_{(u,X)\neq 0}\frac{Q(u,X)}{\|u\|^2 + \frac{1+\Delta t}{2}\|X\|_F^2} \geq \frac{\Delta t}{4+2\Delta t+\Gamma},$$

which is because $g(\nu) := 1 - \nu(4+2\Delta t+\Gamma) + 2(1+\Delta t)\nu^2 \geq 0$ at $\nu = \frac{1}{4+2\Delta t+\Gamma}$. We thus have

$$\|M_{\Delta t,\star}^{\mathrm{KL}}(\xi)\|_{\star,\Delta t} \leq \left(1 - \frac{\Delta t}{4+2\Delta t+\Gamma}\right)\|\xi\|_{\star,\Delta t}$$

since $0 < \Delta t \leq \left(\beta + \frac{1-\alpha}{2}\right)^{-1}$ and hence $\langle M_{\Delta t,\star}^{\mathrm{KL}}(\xi),\xi\rangle_{\star,\Delta t} \geq 0$.

We proceed to control the nonlinear part. Along $a = a_\star + s\xi \in \mathbb{R}^{N_\theta}\times$, we compute

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}_{\rho_{a+s\xi}}[\nabla^2 V]\Big|_{s=0} = T^*[u] + \frac{1}{2}H[X] := A_1(\xi)$$

and bound

$$\|A_1(\xi)\|_F \leq \left(\frac{\beta-\alpha}{2} + \sqrt{2}\beta N_\theta\right)\|\xi\|_\star.$$

From Lemma 3.7,

$$\|D^2F[\zeta,\zeta]\|_{\star,\Delta t} \leq \sqrt{1+\Delta t}\|D^2F[\zeta,\zeta]\|_\star \leq 135\sqrt{1+\Delta t}\beta N_\theta^{5/2}\|\zeta\|_\star^2 \leq 135\sqrt{2}\beta N_\theta^{5/2}\|\zeta\|_{\star,\Delta t}^2.$$

By expanding the covariance update $C_{n+1} = (1+\Delta t)(C_n + \Delta t\mathbb{E}_{\rho_{a_n}}[\nabla^2 V])^{-1}$ and combining the second-order correction bound with continuous-time high-order remainder, we obtain

$$\begin{aligned}\|R_{\Delta t}^{\mathrm{KL}}(\xi_n)\|_{\star,\Delta t} &\leq \frac{\Delta t}{2}\left(135\sqrt{2}\beta N_\theta^{5/2} + 2\sqrt{2} + 8\left(\frac{\beta-\alpha}{2} + \sqrt{2}\beta N_\theta\right) + \sqrt{2}\left(\frac{\beta-\alpha}{2} + \sqrt{2}\beta N_\theta\right)^2\right)\|\xi_n\|_{\star,\Delta t}^2 \\ &\leq \frac{\Delta t}{2}\left(135\sqrt{2}\beta N_\theta^{5/2} + 2\sqrt{2} + 16\beta N_\theta + 4\sqrt{2}\beta^2 N_\theta^2\right)\|\xi_n\|_{\star,\Delta t}^2 \leq \frac{\Delta t}{2}216\beta^2 N_\theta^{5/2}\|\xi_n\|_{\star,\Delta t}^2,\end{aligned}$$

where the vector field component from the first inequality comes from Lemma 3.7 and $\|\xi_n\|_\star \leq \delta_{\mathrm{KL}}(\Delta t) \leq \frac{1}{2\sqrt{2}}$, which ensures that $a_n \in \mathcal{S}_\star$.

Therefore, as the stepsize is chosen to satisfy $0 < \Delta t \leq \left(\beta + \frac{1-\alpha}{2}\right)^{-1}$, then

$$\begin{aligned}\|\xi_{n+1}\|_{\star,\Delta t} &\leq \|M_{\Delta t,\star}^{\mathrm{KL}}(\xi)\|_{\star,\Delta t} + \|R_{\Delta t}^{\mathrm{KL}}(\xi_n)\|_{\star,\Delta t} \\ &\leq \left(1 - \frac{\Delta t}{4+2\Delta t+\Gamma}\right)\|\xi\|_{\star,\Delta t} + \frac{\Delta t}{2}216\beta^2 N_\theta^{5/2}\|\xi_n\|_{\star,\Delta t}^2 \\ &\leq \left(1 - \frac{\Delta t}{2(4+2\Delta t+\Gamma)}\right)\|\xi\|_{\star,\Delta t},\end{aligned}$$

where the last inequality comes from $\|\xi_n\|_{\star,\Delta t} \leq \delta_{\mathrm{KL}}(\Delta t) \leq \frac{1}{(4+2\Delta t+\Gamma)\cdot 216\beta^2 N_\theta^{5/2}}$. This completes the proof by iterating on $n$. □

# 4 General affine invariant Gaussian flows

As Fisher–Rao metric constrained on Gaussian space is affine invariant, we aim to classify all affine-invariant Riemannian metrics on Gaussian space and study the gradient flows they induce for KL divergence.

**Theorem 4.1** (Classification of affine-invariant metrics in Gaussian space)**.** *Every affine-invariant Riemannian metric on the Gaussian manifold $\mathcal{A} = \mathbb{R}^{N_\theta} \times \mathbb{S}_{++}(N_\theta)$ is of the form*

$$g_a((u,X),(v,Y)) = u^\top C^{-1}v + \omega\operatorname{Tr}(C^{-1}XC^{-1}Y) + \tau\operatorname{Tr}(C^{-1}X)\operatorname{Tr}(C^{-1}Y), \qquad \omega > 0, \quad \tau > -\frac{\omega}{N_\theta},$$

*with $a = (m,C) \in \mathcal{A}$ and $(u,X),(v,Y) \in T_a\mathcal{A}$.*

*Proof.* Let $\varphi(\theta) = A\theta + b$ be an invertible affine transformation, we denote the map induced by $\varphi$ as $\Phi_{(A,b)} : \mathcal{A} \to \mathcal{A}$ such that $\Phi_{(A,b)}(m,C) = (Am+b, ACA^\top)$, so that the differential of the map is given by $D\Phi(m,C) : T_a\mathcal{A} \to T_a\mathcal{A}$ with

$$D\Phi(m,C)[(u,X)] = \lim_{\epsilon\to 0}\frac{\Phi(m+\epsilon u, C+\epsilon X) - \Phi(m,C)}{\epsilon} = (Au, AXA^\top),$$

hence affine invariance requires that, for any affine transformation $\varphi$, the metric $g_a : T_a\mathcal{A}\times T_a\mathcal{A} \to \mathbb{R}$ should satisfy

$$g_{(m,C)}((u,X),(v,Y)) = g_{\Phi(m,C)}(D\Phi(m,C)[(u,X)], D\Phi(m,C)[(v,Y)]) = g_{\Phi(m,C)}((Au, AXA^\top),(Av, AYA^\top)).$$

We first state the established result on the classification of all affine invariant Riemannian metrics on $\mathbb{S}_{++}(N_\theta)$ [12, 13]. For any $C \in \mathbb{S}_{++}(N_\theta)$, the family of affine invariant metrics $g_C : T_C\,\mathbb{S}_{++}(N_\theta) \times T_C\,\mathbb{S}_{++}(N_\theta) \to \mathbb{R}$ is characterized by, for any $X, Y \in \mathbb{S}_{++}(N_\theta)$,

$$g_C(X,Y) = \omega\,\mathrm{Tr}(C^{-1}XC^{-1}Y) + \tau\,\mathrm{Tr}(C^{-1}XC^{-1}Y), \qquad \tau > -\frac{\omega}{N_\theta}.$$

We then show that there are no cross terms between $u$ and $Y$ in the affine invariant metrics on Gaussian manifold. We define $L(u,Y) := g_{\mathcal{A}}((u,0),(0,Y))$, for any orthogonal $Q \in O(N_\theta)$, affine invariance requires $L(u,Y) = L(Qu, QYQ^\top)$. We then take $Y = I$, hence $L(u,I) = L(Qu,I)$. Since $Q$ is arbitrary, it forces that $L(u,I) = 0$ for any $u \in \mathbb{R}^{N_\theta}$, which requires no mean-covariance coupling in affine invariant metrics for Gaussian gradient flow.

We proceed to look into the mean part, and first consider the special case on $(0,I)$. Affine invariance requires that for any orthogonal $Q \in O(N_\theta)$, $g_{(0,I)}((Qu,0),(Qv,0)) = g_{(0,I)}((u,0),(v,0))$. Hence, $g_{(0,I)}((u,0),(v,0)) = \eta u^\top v$ for some scalar $\eta \in \mathbb{R}$. We then consider arbitrary covariance matrix $C \in \mathbb{S}_{++}(N_\theta)$, hence $\exists A \in \mathbb{R}^{N_\theta} \times \mathbb{R}^{N_\theta}$ with $C = AA^\top$. Since $\Phi_{(A,m)}(0,I) = (m,C)$, then $g_{(m,C)}((Au,0),(Av,0)) = g_{(0,I)}((u,0),(v,0)) = \eta u^\top v$, which gives $g_{(m,C)}((u',0),(v',0)) = \eta u'^\top C^{-1} v'$ as we write $u' = Au$ and $v' = Av$. Therefore the mean vector part is of form $\eta u^\top C^{-1} v$.

We thus obtain the expression of general affine invariant Riemannian metrics on Gaussian manifold

$$g_a((u,X),(v,Y)) = \eta u^\top C^{-1}v + \omega\,\mathrm{Tr}(C^{-1}XC^{-1}Y) + \tau\,\mathrm{Tr}(C^{-1}X)\,\mathrm{Tr}(C^{-1}Y), \qquad \eta,\,\omega > 0, \quad \tau > -\frac{\omega}{N_\theta},$$

which completes the proof by normalize the coefficient of the first term to 1 to remove irrelevant overall scaling. □

We consider the general affine invariant gradient flow on Gaussian space, where we choose KL divergence as the energy functional following Section 3 of [7] and the affine invariant metrics in Theorem 4.1 as the Riemannian metrics. Thus, we obtain the classification of affine invariant Gaussian gradient flows, which can be characterized by the following ODEs for mean and covariance respectively:

$$\begin{aligned}
\frac{\mathrm{d}m_t}{\mathrm{d}t} &= C_t \mathbb{E}_{\theta\sim\mathcal{N}(m_t,C_t)}[\nabla_\theta \log\rho_{\mathrm{post}}(\theta)],\\
\frac{\mathrm{d}C_t}{\mathrm{d}t} &= \frac{1}{2\omega}(C_t + C_t\mathbb{E}_{\theta\sim\mathcal{N}(m_t,C_t)}[\nabla_\theta\nabla_\theta \log\rho_{\mathrm{post}}(\theta)]C_t)\\
&\quad - \frac{\tau}{2\omega(\omega+\tau N_\theta)}\,\mathrm{Tr}(I + C_t\mathbb{E}_{\theta\sim\mathcal{N}(m_t,C_t)}[\nabla_\theta\nabla_\theta \log\rho_{\mathrm{post}}(\theta)])C_t.
\end{aligned} \tag{10}$$

When $\rho_{\mathrm{post}} \sim \mathcal{N}(m,C)$, without loss of generality, by affine invariance we assume $\rho_{\mathrm{post}} \sim \mathcal{N}(0, I_{N_\theta})$, and hence (10) becomes

$$\begin{aligned}
\frac{\mathrm{d}m_t}{\mathrm{d}t} &= -C_t m_t,\\
\frac{\mathrm{d}C_t}{\mathrm{d}t} &= \frac{1}{2\omega}(C_t - C_t^2) - \frac{\tau}{2\omega(\omega+\tau N_\theta)}\,\mathrm{Tr}(I - C_t)C_t.
\end{aligned}$$

In the covariance part, we write $C = I + X$, and the linearized covariance ODE reads

$$\frac{\mathrm{d}X_t}{\mathrm{d}t} = -\frac{1}{2\omega}X_t + \frac{\tau}{2\omega(\omega+\tau N_\theta)}\operatorname{Tr}(X_t)I + O(\|X_t\|^2).$$

If we furthermore split $X$ into diagonal and off-diagonal part, i.e., $X = X_0 + \frac{\operatorname{Tr}(X)}{N_\theta}I$, then

$$\dot{X}_0 = -\frac{1}{2\omega}X_0 + O(\|X\|^2), \qquad \frac{\mathrm{d}}{\mathrm{d}t}\operatorname{Tr}(X_t) = -\frac{1}{2(\omega+\tau N_\theta)}\operatorname{Tr}(X_t) + O(\|X_t\|^2).$$

The discretization with Riemannian distance reads

$$\begin{aligned}
&m_{n+1} = m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\mathrm{post}}(\theta)],\\
&C_{n+1} = C_n^{1/2}\exp\left(\frac{\Delta t}{2\omega}\left[C_n^{1/2}\mathbb{E}_{\rho_{a_n}}[\nabla_\theta\nabla_\theta \log\rho_{\mathrm{post}}(\theta)]C_n^{1/2} + \frac{\omega - \tau\operatorname{Tr}(C_n\mathbb{E}_{\rho_{a_n}}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}(\theta)])}{\omega+N_\theta\tau}I\right]\right)C_n^{1/2},
\end{aligned} \tag{11}$$

and in the special case where $\omega = \frac{1}{2}, \tau = 0$, we recover (2).

Based on the Gaussian natural gradient discretization with KL divergence (4), we write down the following discretization scheme

$$\begin{aligned}
&m_{n+1} = m_n + \Delta t C_n \mathbb{E}_{\rho_{a_n}}[\nabla_\theta \log \rho_{\mathrm{post}}(\theta)],\\
&C_{n+1} = \left[1 + \frac{\Delta t}{2\omega(\omega+\tau N_\theta)}(\omega - \tau\operatorname{Tr}(C_n\mathbb{E}_{\rho_{a_n}}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}(\theta)]))\right]\left(C_n^{-1} - \frac{\Delta t}{2\omega}\mathbb{E}_{\rho_{a_n}}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}(\theta)]\right)^{-1}.
\end{aligned} \tag{12}$$

This update recovers (4) when $\omega = \frac{1}{2}, \tau = 0$. As the scheme is first-order consistent with the ODE (10), but it's not the exact update of a proximal algorithm.

# 5 Wasserstein–Fisher–Rao gradient flow on Gaussian manifold

Let $\psi_\rho = \frac{\delta\mathcal{E}}{\delta\rho} = \log\rho - \log\rho_{\mathrm{post}}$, we consider using the following transport-accelerated Fisher–Rao gradient flow:

$$\partial_t\rho_t = -\rho_t(\psi_{\rho_t} - \mathbb{E}_{\rho_t}[\psi_{\rho_t}]) + \lambda_t\nabla\cdot(\rho_t\nabla\psi_{\rho_t}),$$

where $\lambda_t \geq 0$ is adaptive, which measures the strength in optimal transport, and recovers standard Fisher–Rao gradient flow when $\lambda_t = 0$. By moment closure as in [7], we obtain the following Wasserstein–Fisher–Rao gradient flow constrained on Gaussian manifold, which can be viewed as transport-accelerated natural gradient:

$$\begin{aligned}
\frac{\mathrm{d}m_t}{\mathrm{d}t} &= (C_t + \lambda_t I)\mathbb{E}_{\rho_{a_t}}[\nabla_\theta\log\rho_{\mathrm{post}}(\theta)],\\
\frac{\mathrm{d}C_t}{\mathrm{d}t} &= C_t + C_t\mathbb{E}_{\rho_{a_t}}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}(\theta)]C_t\\
&\quad + \lambda_t(2I + C_t\mathbb{E}_{\rho_{a_t}}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}(\theta)] + \mathbb{E}_{\rho_{a_t}}[\nabla_\theta\nabla_\theta\log\rho_{\mathrm{post}}(\theta)]C_t).
\end{aligned} \tag{13}$$

We prove the convergence rate of (13) in the following Theorem.

**Theorem 5.1.** *Under Assumption 2.1 and assume that the spectral bounds of the initial covariance $\mu_{0,\min}I \preceq C_0 \preceq \mu_{0,\max}I$, for the Wasserstein–Fisher–Rao gradient flow on Gaussian space (13) with locally integrable adaptive schedule $\lambda_t$, we have the following convergence guarantee:*

$$\mathcal{E}(a_t) - \mathcal{E}(a_\star) \leq \exp\left(-2\alpha t\min\left\{\mu_{0,\min}, \frac{1}{\beta}\right\} - 2\alpha\int_0^t \lambda_s \mathrm{d}s\right)(\mathcal{E}(a_0) - \mathcal{E}(a_\star)).$$

*Proof.* We first prove the covariance bound along (13). Let $\mu_i(t)$ be the $i$-th eigenvalue of $C_t$ with corresponding normalized eigenvector $u_i(t)$, hence

$$\begin{aligned}\frac{\mathrm{d}\mu_i(t)}{\mathrm{d}t} &= u_i(t)^\top \frac{\mathrm{d}C_t}{\mathrm{d}t} u_i(t) \\ &= \mu_i - \mu_i^2 h_i + 2\lambda_t(1 - \mu_i h_i) \\ &= (\mu_i(t) + 2\lambda_t)(1 - \mu_i(t) h_i(t))\end{aligned}$$

where $h_i(t) = -u_i(t)^\top \mathbb{E}_{\rho_{a_t}}[\nabla_\theta \nabla_\theta \log \rho_{\mathrm{post}}(\theta)] u_i(t) \in [\alpha, \beta]$ since Assumption 2.1. Because $\mu_i(t) \geq 0$ and $h_i(t) \geq 0$, then if $\mu_i(t) \geq \frac{1}{\alpha}$, $\frac{\mathrm{d}\mu_i(t)}{\mathrm{d}t} \leq 0$; if $\mu_i(t) \leq \frac{1}{\beta}$, $\frac{\mathrm{d}\mu_i(t)}{\mathrm{d}t} \geq 0$. This gives the covariance spectral bound $\mu_{\min} I \preceq C_t \preceq \mu_{\max} I$ with

$$\mu_{\min} := \min\left\{\mu_{0,\min}, \frac{1}{\beta}\right\}, \qquad \mu_{\max} := \max\left\{\mu_{0,\max}, \frac{1}{\alpha}\right\}.$$

Along the Gaussian approximate Wasserstein–Fisher–Rao gradient flow (13), we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}(a_t) &= -\|\operatorname{grad}\mathcal{E}(a_t)\|_{a_t}^2 - \lambda_t \|\operatorname{grad}^{W_2}\mathcal{E}(a_t)\|_{W_2}^2 \\ &\leq -(\mu_{\min} + \lambda_t)\|\operatorname{grad}^{W_2}\mathcal{E}(a_t)\|_{W_2}^2 \\ &\leq -2\alpha(\mu_{\min} + \lambda_t)(\mathcal{E}(a_t) - \mathcal{E}(a_\star)),\end{aligned}$$

where the first inequality connects Fisher–Rao geometry to Wasserstein geometry following the technique as in the proof of Theorem 2.3, while the second inequality follows the PL-inequality in Wasserstein geometry as $\mathcal{E}$ is $\alpha$-strongly convex along $W_2$ geodesics [10]. We thus obtain the final convergence guarantee following Gronwall's inequality:

$$\mathcal{E}(a_t) - \mathcal{E}(a_\star) \leq \exp\left(-2\alpha\mu_{\min} t - 2\alpha \int_0^t \lambda_s \mathrm{d}s\right)(\mathcal{E}(a_0) - \mathcal{E}(a_\star)).$$

□

We propose a forward-backward algorithm (Algorithm 1) to discretize (13), where the Wasserstein part comes from Algorithm 1 of [8] while the Fisher–Rao part comes from (4), and prove its convergence rate in Theorem 5.2.

**Theorem 5.2.** *Under Assumption 2.1 and assume that the initial covariance has spectral bounds $\mu_{0,\min} I \preceq C_0 \preceq \mu_{0,\max} I$, if we denote $\mu_{\min} := \min\left\{\mu_{0,\min}, \frac{1}{\beta}\right\}$, $\mu_{\max} := \max\left\{\mu_{0,\max}, \frac{1}{\alpha}\right\}$, $L = \beta\mu_{\max}\max\left\{1, \frac{\mu_{\max}^3}{2\mu_{\min}^3}\right\}$, and choose stepsize satisfying $0 < \Delta t \leq \frac{1}{\max\{L, \beta \max_{0 \leq n \leq N-1} \lambda_n\}}$, the Algorithm 1 has the following convergence guarantee:*

$$\begin{aligned}\mathcal{E}(a_N) - \mathcal{E}(a_\star) \leq \exp\Bigg(&-2\alpha\mu_{\min} N\Delta t \min\left\{\frac{1 + \Delta t \alpha \mu_{\min}}{2(1 + \Delta t)}, \frac{1}{4(1 + \Delta t \beta \mu_{\max})^2}\right\} \\ &- \sum_{n=0}^{N-1} \frac{\alpha \lambda_n \Delta t}{(1 + \lambda_n \Delta t / \mu_{\min})^2}\Bigg)(\mathcal{E}(a_0) - \mathcal{E}(a_\star)).\end{aligned}$$

*Proof.* We first show that the spectral bounds of covariance along Algorithm 1 is given by $\mu_{\min} I \preceq C \preceq \mu_{\max} I$. We assume inductively that $\mu_{\min} I \preceq C_n \preceq \mu_{\max} I$. For the Wasserstein update,

$$(1 - \lambda_n \Delta t \beta) I \preceq M = I + \lambda_n \Delta t H_n \preceq (1 - \lambda_n \Delta t \alpha) I,$$

**Algorithm 1** Forward–backward algorithm of Gaussian approximate Wasserstein–Fisher–Rao gradient flow

**Require:** Stepsize $\Delta t > 0$; iteration count $N$; initial Gaussian parameter $a_0 = (m_0, C_0)$ with $C_0 \in \mathbb{S}_{++}^{N_\theta}$; adaptive rule $\lambda_n \geq 0$

**for** $n = 0, \dots, N-1$ **do**

Compute the Gaussian expectations at $a_n = (m_n, C_n)$:

$$g_n \leftarrow \mathbb{E}_{\rho_{a_n}}\big[\nabla_\theta \log \rho_{\text{post}}(\theta)\big], \qquad H_n \leftarrow \mathbb{E}_{\rho_{a_n}}\big[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)\big].$$

**Wasserstein forward–backward step:**

$$m_{n+1/2} \leftarrow m_n + \lambda_n \Delta t g_n, \quad M_n \leftarrow I + \lambda_n \Delta t H_n, \quad \widetilde{C}_{n+\frac{1}{2}} \leftarrow M_n C_n M_n,$$
$$C_{n+1/2} \leftarrow \frac{1}{2}\left(\widetilde{C}_{n+1/2} + 2\lambda_n \Delta t I + \left[\widetilde{C}_{n+1/2}\left(\widetilde{C}_{n+1/2} + 4\lambda_n \Delta t I\right)\right]^{1/2}\right).$$

Set $a_{n+1/2} \leftarrow (m_{n+1/2}, C_{n+1/2})$ and compute the Gaussian expectations at $a_{n+1/2}$:

$$g_{n+1/2} \leftarrow \mathbb{E}_{\rho_{a_{n+1/2}}}\big[\nabla_\theta \log \rho_{\text{post}}(\theta)\big], \qquad H_{n+1/2} \leftarrow \mathbb{E}_{\rho_{a_{n+1/2}}}\big[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)\big].$$

**Fisher–Rao forward–backward step:**

$$m_{n+1} \leftarrow m_{n+1/2} + \Delta t\, C_{n+1/2}\, g_{n+1/2}, \qquad C_{n+1} \leftarrow (1+\Delta t)\left(C_{n+1/2}^{-1} - \Delta t\, H_{n+1/2}\right)^{-1}.$$

Set $a_{n+1} \leftarrow (m_{n+1}, C_{n+1})$.

**end for**

**return** $a_N = (m_N, C_N)$ and $\rho_{a_N} = \mathcal{N}(m_N, C_N)$.

hence

$$(1 - \lambda_n \Delta t \beta)^2 \mu_{\min} I \preceq \widetilde{C}_{n+1/2} \preceq (1 - \lambda_n \Delta t \alpha)^2 \mu_{\max} I.$$

The backward step of covariance is characterized by the following update map

$$\Psi_{\Delta t}(X) = \frac{1}{2}(X + 2\lambda_n \Delta t I + [X(X + 4\lambda_n \Delta t I)]^{1/2}),$$

on eigenvalues, the corresponding map is

$$\psi_{\Delta t}(x) = \frac{1}{2}\left(x + 2\lambda_n \Delta t + \sqrt{x(x + 4\lambda_n \Delta t)}\right),$$

which is monotonically increasing and we can calculate the following identity

$$\psi_{\Delta t}\left(\left(1 - \frac{\lambda_n \Delta t}{c}\right)^2 c\right) = c, \qquad 0 \leq \lambda_n \Delta t \leq c.$$

We first investigate the lower bound. If $\lambda_n \Delta t \geq \mu_{\min}$, then

$$\psi_{\Delta t}(x) \geq \psi_{\Delta t}(0) = \lambda_n \Delta t \geq \mu_{\min},$$

if $\lambda_n \Delta t < \mu_{\min}$, since $\mu_{\min} \leq \frac{1}{\beta}$, then

$$\psi_{\Delta t}((1 - \lambda_n \Delta t \beta)^2 \mu_{\min}) \geq \psi_{\Delta t}\left(\left(1 - \frac{\lambda_n \Delta t}{\mu_{\min}}\right)^2 \mu_{\min}\right) = \mu_{\min},$$

these collectively give $C_{n+1/2} \succeq \mu_{\min} I$. For the upper bound, since $\mu_{\max} \geq \frac{1}{\alpha}$, then

$$\psi_{\Delta t}((1-\lambda_n \Delta t \alpha)^2 \mu_{\max}) \leq \psi_{\Delta t}\left(\left(1-\frac{\lambda_n \Delta t}{\mu_{\max}}\right)^2 \mu_{\max}\right) = \mu_{\max},$$

hence $C_{n+1/2} \preceq \mu_{\max} I$ and hence $\mu_{\min} I \preceq C_{n+1/2} \preceq \mu_{\max} I$. By Lemma 2.7, we have $\mu_{\min} I \preceq C_{n+1} \preceq \mu_{\max} I$ for the Fisher–Rao step, which completes the covariance bounds proof.

We proceed to investigate the evolution from $a_n$ to $a_{n+1/2}$. If $\lambda_n = 0$, it's trivial that $a_{n+1/2} = a_n$, so we consider the case where $\lambda_n > 0$. We choose $\nu = a_n$ in Lemma 5.1 of [8] to obtain the following inequality characterizing the Wasserstein update:

$$\mathcal{E}(a_n) - \mathcal{E}(a_{n+1/2}) \geq \frac{1}{2\lambda_n \Delta t} W_2^2(\rho_{a_n}, \rho_{a_{n+1/2}}) = \frac{\lambda_n \Delta t}{2} \|G(a_n)\|_{L^2(\rho_{a_n})}^2,$$

where we denote $G(a_n) := \frac{I - T_n}{\lambda_n \Delta t}$ and $T_n$ is the optimal transport map from $a_n$ to $a_{n+1/2}$. The optimality condition of the Wasserstein forward-backward step gives

$$G(a_n) = \operatorname{grad}^{W_2} \mathcal{E}_2(a_n) + \operatorname{grad}^{W_2} \mathcal{E}_1(a_{n+1/2}) \circ T_n.$$

Since $\operatorname{grad}^{W_2} \mathcal{E}(a_n) = \operatorname{grad}^{W_2} \mathcal{E}_1(a_n) + \operatorname{grad}^{W_2} \mathcal{E}_2(a_n)$, then by the covariance spectral bounds, we have

$$\begin{aligned}
\|\operatorname{grad}^{W_2} \mathcal{E}(a_n) - G(a_n)\|_{L^2(\rho_{a_n})} &= \|\operatorname{grad}^{W_2} \mathcal{E}_1(a_n) - \operatorname{grad}^{W_2} \mathcal{E}_1(a_{n+1/2}) \circ T_n\|_{L^2(\rho_{a_n})} \\
&= \|\nabla_\theta \log \rho_{a_n} - \nabla_\theta \log \rho_{a_{n+1/2}} \circ T_n\|_{L^2(\rho_{a_n})} \\
&\leq \frac{1}{\mu_{\min}} \|I - T_n\|_{L^2(\rho_{a_n})} \leq \frac{\lambda_n \Delta t}{\mu_{\min}} \|G(a_n)\|_{L^2(\rho_{a_n})}.
\end{aligned}$$

Therefore,

$$\|\operatorname{grad}^{W_2} \mathcal{E}(a_n)\|_{W_2} \leq \left(1 + \frac{\lambda_n \Delta t}{\mu_{\min}}\right) \|G(a_n)\|_{L^2(\rho_{a_n})},$$

then by PL inequality on Wasserstein geometry [10],

$$\mathcal{E}(a_n) - \mathcal{E}(a_{n+1/2}) \geq \frac{\lambda_n \Delta t}{2(1+\lambda_n \Delta t/\mu_{\min})^2} \|\operatorname{grad}^{W_2} \mathcal{E}(a_n)\|_{W_2}^2 \geq \frac{\alpha \lambda_n \Delta t}{(1+\lambda_n \Delta t/\mu_{\min})^2} (\mathcal{E}(a_n) - \mathcal{E}(a_\star)).$$

This gives the following one-step inequality of Wasserstein part

$$\mathcal{E}(a_{n+1/2}) - \mathcal{E}(a_\star) \leq \left[1 - \frac{\alpha \lambda_n \Delta t}{(1+\lambda_n \Delta t/\mu_{\min})^2}\right] (\mathcal{E}(a_n) - \mathcal{E}(a_\star)).$$

By Theorem 2.10, the one-step inequality of Fisher–Rao part reads

$$\mathcal{E}(a_{n+1}) - \mathcal{E}(a_\star) \leq \left[1 - 2\alpha \mu_{\min} \Delta t \min\left\{\frac{1+\Delta t \alpha \mu_{\min}}{2(1+\Delta t)}, \frac{1}{4(1+\Delta t \beta \mu_{\max})^2}\right\}\right] (\mathcal{E}(a_{n+1/2}) - \mathcal{E}(a_\star)).$$

We iterate on $n$ and apply $1 - x \leq e^{-x}$ to obtain the convergence rate of Algorithm 1:

$$\begin{aligned}
\mathcal{E}(a_N) - \mathcal{E}(a_\star) \leq \exp\Bigg(&-2\alpha \mu_{\min} N \Delta t \min\left\{\frac{1+\Delta t \alpha \mu_{\min}}{2(1+\Delta t)}, \frac{1}{4(1+\Delta t \beta \mu_{\max})^2}\right\} \\
&- \sum_{n=0}^{N-1} \frac{\alpha \lambda_n \Delta t}{(1+\lambda_n \Delta t/\mu_{\min})^2}\Bigg) (\mathcal{E}(a_0) - \mathcal{E}(a_\star)).
\end{aligned}$$

□

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